Centre National de la Recherche Scientifique

Complete list of publications

Laurent U Perrinet



Team NEural OPerations in TOpographies (NeOpTo)
Institut de Neurosciences de la Timone UMR 7289, CNRS / Aix-Marseille University
27, Bd. Jean Moulin, 13385 Marseille Cedex 5, France
https://laurentperrinet.github.io/
Laurent.Perrinet@univ-amu.fr

24th October 2026

1 Journal articles under review

A65 Giacomo Benvenuti, Sandrine Chemla, Arjan Boonman, Laurent U Perrinet, Guillaume S Masson and Frédéric Y Chavane. 'Anticipatory Responses along Motion Trajectories in Awake Monkey Area V1'. en. In: *bioRxiv* (2020), p. 2020.03.26.010017. DOI: 10.1101/2020.03.26.010017. URL: https://www.biorxiv.org/content/10.1101/2020.03.26.010017

A64 Laurent U Perrinet. 'Working Memory with Polychronous Chains'. In: *arXiv preprint arXiv:2604.14096* (2026). DOI: 10.48550/arXiv.2604.14096. URL: https://laurentperrinet.github.io/publication/perrinet-26

A63 Matthis Dallain, Laurent Rodriguez, Laurent U Perrinet and Benoît Miramond. 'A Saccade-Inspired Approach to Image Classification Using Visiontransformer Attention Maps'. working paper. 10th Mar. 2026. DOI: 10.48550/arXiv.2603.09613. arXiv: 2603.09613 [cs]. URL: http://arxiv.org/abs/2603.09613 (visited on 11/03/2026). Pre-published

2 Peer-reviewed international journal articles

2026

A62 Kevin Mairot, Isabelle Meunier, Béatrice Bocquet, Natacha Stolowy, Prithvi Ramtohul, Xavier Zanlonghi, Danièle Denis, Thierry David, Frederic Matonti and Laurent Udo Perrinet. 'Deep Learning for Diagnosis of Choroideremia and USH2A-Associated RodCone Dystrophy Using Macular Optical Coherence Tomography Volumes'. In: *Ophthalmology Science* 6.10 (Oct. 2026), p. 101378. ISSN: 2666-9145. DOI: 10.1016/j.xops.2026.101378. URL: https://doi.org/10.1016/j.xops.2026.101378 (visited on 27/08/2026)

A61 Thomas Kronland-Martinet, Stéphane Viollet and Laurent U Perrinet. 'Detection of spiking motifs of arbitrary length in neural activity using bounded synaptic delays'. In: *Biological Cybernetics* 120.5–6 (17th Aug. 2026), p. 27. ISSN: 1432-0770. DOI: 10.1007/s00422-026-01051-7

A60 Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U. Perrinet. 'Foveated Retinotopy Improves Classification and Localization in Convolutional Neural Networks'. In: *Vision* 10.2 (June 2026), p. 17. ISSN: 2411-5150. DOI: 10.3390/vision10020017. URL: https://www.mdpi.com/2411-5150/10/2/17

2025

A59 Andrew Isaac Meso, Jonathan Vacher, Nikos Gekas, Pascal Mamassian, Laurent U Perrinet and Guillaume S Masson. 'DynTex: A Real-Time Generative Model of Dynamic Naturalistic Luminance Textures'. In: *Journal of Vision* 25.11 (Sept. 2025), p. 2. ISSN: 1534-7362. DOI: 10.1167/jov.25.11.2. URL: https://doi.org/10.1167/jov.25.11.2 (visited on 03/09/2025)

A58 Skye Gunasekaran, Assel Kembay, Hugo Ladret, Rui-Jie Zhu, Laurent U Perrinet, Omid Kavehei and Jason Eshraghian. 'A Predictive Approach to Enhance Time-Series Forecasting'. In: *Nature Communications* (19th Oct. 2025). DOI: 10.1038/s41467-025-63786-4. arXiv: 2410.15217. URL: https://www.nature.com/articles/s41467-025-63786-4

2024

A57 Antoine Grimaldi, Victor Boutin, Sio-Hoi Ieng, Ryad Benosman and Laurent U Perrinet. 'A Robust Event-Driven Approach to Always-on Object Recognition'. In: *Neural Networks* 178 (1st Oct. 2024), p. 106415. DOI: 10.1016/j.neunet.2024.106415. URL: https://laurentperrinet.github.io/publication/grimaldi-24/

A56 Hugo Ladret, Christian Casanova and Laurent U Perrinet. 'Kernel Heterogeneity Improves Sparseness of Natural Images Representations'. In: *Neuromorphic Computing and Engineering* (20th Aug. 2024). DOI: 10.1088/2634-4386/ad5d0f. URL: https://iopscience.iop.org/article/10.1088/2634-4386/ad5d0f

2023

A55 Ilias Rentzeperis, Luca Calatroni, Laurent U Perrinet and Dario Prandi. 'Beyond ℓ_1 sparse coding in V1'. In: *PLOS Computational Biology* (12th Sept. 2023). DOI: 10.1371/journal.pcbi.1011459. URL: https://laurentperrinet.github.io/publication/rentzeperis-23

A54 Amélie Gruel, Dalia Hareb, Antoine Grimaldi, Jean Martinet, Laurent U Perrinet, Bernabé Linares-Barranco and Teresa Serrano-Gotarredona. 'Stakes of Neuromorphic Foveation: a promising future for embedded event cameras'. In: *Biological Cybernetics* (2023). DOI: 10.1007/s00422-023-00974-9

A53 Antoine Grimaldi and Laurent U Perrinet. 'Learning heterogeneous delays in a layer of spiking neurons for fast motion detection'. In: *Biological Cybernetics* (11th Sept. 2023). DOI: 10.1007/s00422-023-00975-8. URL: https://laurentperrinet.github.io/publication/grimaldi-23-bc/

A52 Hugo Ladret, Nelson Cortes, Lamyae Ikan, Frédéric Chavane, Christian Casanova and Laurent U Perrinet. 'Cortical recurrence supports resilience to sensory variance in the primary visual cortex'. In: *Nature Communications Biology* (6th June 2023). DOI: 10.1038/s42003-023-05042-3. URL: https://www.nature.com/articles/s42003-023-05042-3

A51 Jean-Nicolas Jérémie and Laurent U Perrinet. 'Ultra-Fast Image Categorization in biology and in neural models'. In: *Vision* (21st Mar. 2023). DOI: 10.3390/vision7020029

2022

A50 Antoine Grimaldi, Amélie Gruel, Camille Besnainou, Jean-Nicolas Jérémie, Jean Martinet and Laurent U Perrinet. 'Precise spiking motifs in neurobiological and neuromorphic data'. In: *Brain Sciences* (23rd Dec. 2022). DOI: 10.3390/brainsci13010068. URL: https://laurentperrinet.github.io/publication/grimaldi-22-polychronies/

A49 Victor Boutin, Angelo Franciosini, Frédéric Y Chavane and Laurent U Perrinet. 'Pooling in a predictive model of V1 explains functional and structural diversity across species'. In: *PLoS Computational Biology* (18th July 2022). DOI: 10.1371/journal.pcbi.1010270. URL: https://laurentperrinet.github.io/publication/franciosini-21

A48 Frédéric V Barthélemy, Jérome Fleuriet, Laurent U Perrinet and Guillaume S Masson. 'A Behavioral Receptive Field for Ocular Following in Monkeys: Spatial Summation and Its Spatial Frequency Tuning'. In: *eNeuro* (2022), ENEURO.0374–21.2022. ISSN: 2373-2822. DOI: 10.1523/ENEURO.0374-21.2022

A47 Frédéric Chavane, Laurent U Perrinet and James Rankin. 'Revisiting Horizontal Connectivity Rules in V1: From like-to-like towards like-to-All'. In: *Brain Structure and Function* (5th Feb. 2022). ISSN: 1863-2661. DOI: 10.1007/s00429-022-02455-4. URL: https://doi.org/10.1007/s00429-022-02455-4

2021

A46 Victor Boutin, Angelo Franciosini, Frédéric Y Chavane, Franck Ruffier and Laurent U Perrinet. 'Sparse Deep Predictive Coding captures contour integration capabilities of the early visual system'. In: *PLoS Computational Biology* (26th Jan. 2021). DOI: 10.1371/journal.pcbi.1008629. URL: https://doi.org/10.1371/journal.pcbi.1008629

2020

A45 Victor Boutin, Angelo Franciosini, Franck Ruffier and Laurent U Perrinet. 'Effect of top-down connections in Hierarchical Sparse Coding'. In: *Neural Computation* 32.11 (4th Feb. 2020), pp. 2279–2309. DOI: 10.1162/neco_a_01325. URL: https://laurentperrinet.github.io/publication/boutin-franciosini-ruffier-perrinet-20-feedback/

A44 Emmanuel Daucé, Pierre Albigès and Laurent U Perrinet. 'A dual foveal-peripheral visual processing model implements efficient saccade selection'. In: *Journal of Vision* 20.8 (5th June 2020), pp. 22–22. DOI: 10.1167/jov.20.8.22. URL: https://laurentperrinet.github.io/publication/dauce-20/

A43 Chloé Pasturel, Anna Montagnini and Laurent U Perrinet. 'Humans adapt their anticipatory eye movements to the volatility of visual motion properties'. In: *PLoS Computational Biology* (26th Jan. 2020). DOI: 10.1371/journal.pcbi.1007438. URL: https://hal.science/hal-02394142

2019

A42 Laurent U Perrinet. 'An adaptive homeostatic algorithm for the unsupervised learning of visual features'. In: *Vision* 3.3 (2019), p. 47. DOI: 10.3390/vision3030047. URL: https://spikeai.github.io/HULK/

A41 Cesar U Ravello, Laurent U Perrinet, Maria-José Escobar and Adrián G Palacios. 'Speed-Selectivity in Retinal Ganglion Cells is Sharpened by Broad Spatial Frequency, Naturalistic Stimuli'. In: *Scientific Reports* 9.1 (24th Jan. 2019). DOI: 10.1038/s41598-018-36861-8. URL: https://doi.org/10.1038%2Fs41598-018-36861-8

A40 Sandrine Chemla, Alexandre Reynaud, Matteo diVolo, Yann Zerlaut, Laurent U Perrinet, Alain Destexhe and Frédéric Y Chavane. 'Suppressive waves disambiguate the representation of long-range apparent motion in awake monkey V1'. In: *Journal of Neuroscience* 2792 (18th Mar. 2019), p. 18. DOI: 10.1523/JNEUROSCI.2792-18.2019. URL: https://www.jneurosci.org/content/39/22/4282

2018

A39 Jean-Bernard Damasse, Laurent U Perrinet, Laurent Madelain and Anna Montagnini. 'Reinforcement effects in anticipatory smooth eye movements'. In: *Journal of Vision* 18.11 (1st Oct. 2018), pp. 14–14. ISSN: 1534-7362. DOI: 10.1167/18.11.14. URL: https://jov.arvojournals.org/article.aspx?articleid=2707670

A38 Jonathan Vacher, Andrew Isaac Meso, Laurent U Perrinet and Gabriel Peyré. 'Bayesian Modeling of Motion Perception using Dynamical Stochastic Textures'. In: *Neural Computation* (21st Nov. 2018). DOI: 10.1162/neco_a_01142. URL: https://www.mitpressjournals.org/doi/abs/10.1162/neco_a_01142

2017

A37 Mina A Khoei, Guillaume S Masson and Laurent U Perrinet. 'The flash-lag effect as a motion-based predictive shift'. In: *PLoS Computational Biology* 13.1 (26th Jan. 2017), e1005068. DOI: 10.1371/journal.pcbi.1005068. URL: https://laurentperrinet.github.io/publication/khoei-masson-perrinet-17/

2016

A36 Jens Kremkow, Laurent U Perrinet, Cyril Monier, Jose-Manuel Alonso, Ad M Aertsen, Yves Frégnac and Guillaume S Masson. 'Push-Pull Receptive Field Organization and Synaptic Depression: Mechanisms for Reliably Encoding Naturalistic Stimuli in V1'. In: *Frontiers in Neural Circuits* 10 (2016). ISSN: 1662-5110. DOI: 10.3389/fncir.2016.00037. URL: http://journal.frontiersin.org/article/10.3389/fncir.2016.00037/full

2015

A36 Wahiba Taouali, Giacomo Benvenuti, Pascal Wallisch, Frédéric Y Chavane and Laurent U Perrinet. 'Testing the odds of inherent vs. observed overdispersion in neural spike counts'. In: *Journal of Neurophysiology* 115.1 (22nd Jan. 2016), pp. 434–444. ISSN: 1522-1598. DOI: 10.1152/jn.00194.2015. URL: http://www.ncbi.nlm.nih.gov/pubmed/26445864

A35 Jonathan Vacher, Andrew Isaac Meso, Laurent U Perrinet and Gabriel Peyré. 'Biologically Inspired Dynamic Textures for Probing Motion Perception'. In: *Advances in Neural Information Processing Systems* 28 (2015), pp. 1918–1926. URL: http://papers.nips.cc/paper/5769-biologically-inspired-dynamic-textures-for-probing-motion-perception.pdf

A34 Laurent U Perrinet and James A Bednar. 'Edge co-occurrences can account for rapid categorization of natural versus animal images'. In: *Scientific Reports* 5 (2015), p. 11400. DOI: 10.1038/srep11400. URL: http://www.nature.com/articles/srep11400

2013

A33 Laurent U Perrinet, Rick A Adams and Karl Friston. 'Active inference, eye movements and oculomotor delays'. In: *Biological Cybernetics* 108.6 (16th Dec. 2014), pp. 777–801. ISSN: 1432-0770. DOI: 10.1007/s00422-014-0620-8

2013

A32 Mina A Khoei, Guillaume S Masson and Laurent U Perrinet. 'Motion-based prediction explains the role of tracking in motion extrapolation'. In: *Journal of Physiology-Paris* 107.5 (1st Nov. 2013), pp. 409–420. ISSN: 0928-4257. DOI: 10.1016/j.jphysparis.2013.08.001. URL: https://laurentperrinet.github.io/publication/khoei-13-jpp/

A31 Bernhard A Kaplan, Anders Lansner, Guillaume S Masson and Laurent U Perrinet. 'Anisotropic connectivity implements motion-based prediction in a spiking neural network'. In: *Frontiers in Computational Neuroscience* 7.112 (17th Sept. 2013). DOI: 10.3389/fncom.2013.00112. URL: https://laurentperrinet.github.io/publication/kaplan-13

A30 Rodrigo Nava, J Victor Marcos, Boris Escalante-Ramirez, Gabriel Cristóbal, Laurent U Perrinet and Raúl S J Estépar. 'Advances in Texture Analysis for Emphysema Classification'. In: *Lecture Notes in Computer Science* 8259 (2013). Ed. by David Hutchison et al., pp. 214–221. ISSN: 1611-3349. DOI: 10.1007/978-3-642-41827-3_27

2012

A29 Claudio Simoncini, Laurent U Perrinet, Anna Montagnini, Pascal Mamassian and Guillaume S Masson. 'More is not always better: dissociation between perception and action explained by adaptive gain control'. In: *Nature Neuroscience* (2012). DOI: 10.1038/nn.3229. URL: http://www.nature.com/neuro/journal/vaop/ncurrent/full/nn.3229.html

A28 Laurent U Perrinet and Guillaume S Masson. 'Motion-based prediction is sufficient to solve the aperture problem'. In: *Neural Computation* 24.10 (2012), pp. 2726–50. DOI: 10.1162/NECO_a_00332

A27 Paula S Leon, Ivo Vanzetta, Guillaume S Masson and Laurent U Perrinet. 'Motion Clouds: Model-based stimulus synthesis of natural-like random textures for the study of motion perception'. In: *Journal of Neurophysiology* 107.11 (14th Mar. 2012), pp. 3217–3226. ISSN: 1522-1598. DOI: 10.1152/jn.00737.2011. URL: http://dx.doi.org/10.1152/jn.00737.2011

A26 Karl Friston, Rick A Adams, Laurent U Perrinet and Michael Breakspear. 'Perceptions as Hypotheses: Saccades as Experiments'. In: *Frontiers in Psychology* 3 (2012). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2012.00151. URL: http://dx.doi.org/10.3389/fpsyg.2012.00151

A25 Rick A Adams, Laurent U Perrinet and Karl Friston. 'Smooth Pursuit and Visual Occlusion: Active Inference and Oculomotor Control in Schizophrenia'. In: *PLoS ONE* 7.10 (26th Oct. 2012), e47502+. DOI: 10.1371/journal.pone.0047502. URL: http://dx.doi.org/10.1371/journal.pone.0047502

A24 Guillaume S Masson and Laurent U Perrinet. 'The behavioral receptive field underlying motion integration for primate tracking eye movements'. In: *Neuroscience and biobehavioral reviews* (21st Mar. 2012). ISSN: 1873-7528. DOI: 10.1016/j.neubiorev.2011.03.009. URL: http://view.ncbi.nlm.nih.gov/pubmed/21421006

A23 Nicole Voges and Laurent U Perrinet. 'Complex dynamics in recurrent cortical networks based on spatially realistic connectivities'. In: *Frontiers in Computational*

Neuroscience 6 (2012). ISSN: 1662-5188. DOI: 10.3389/fncom.2012.00041. URL: https://laurentperrinet.github.io/publication/voges-12

2011

A22 Jérôme Fleuriet, Sandrine Hugues, Laurent U Perrinet and Laurent Goffart. 'Saccadic foveation of a moving visual target in the rhesus monkey'. In: *Journal of Neurophysiology* 105.2 (1st Feb. 2011), pp. 883–895. ISSN: 1522-1598. DOI: 10.1152/jn.00622.2010. URL: http://dx.doi.org/10.1152/jn.00622.2010

A21 Amarender Bogadhi, Anna Montagnini, Pascal Mamassian, Laurent U Perrinet and Guillaume S Masson. 'Pursuing motion illusions: a realistic oculomotor framework for Bayesian inference'. In: *Vision research* 51.8 (22nd Apr. 2011), pp. 867–880. ISSN: 1878-5646. DOI: 10.1016/j.visres.2010.10.021. URL: http://dx.doi.org/10.1016/j.visres.2010.10.021

2010

A20 Laurent U Perrinet. 'Role of homeostasis in learning sparse representations'. In: *Neural Computation* 22.7 (17th July 2010), pp. 1812–36. ISSN: 1530-888X. DOI: 10.1162/neco.2010.05-08-795. URL: https://doi.org/10.1162/neco.2010.05-08-795

A19 Emmanuel Daucé and Laurent U Perrinet. 'Computational Neuroscience, from Multiple Levels to Multi-level'. In: *Journal of Physiology-Paris* 104.1–2 (2010), pp. 1–4. DOI: 10.1016/j.jphysparis.2009.11.001. URL: http://dx.doi.org/10.1016/j.jphysparis.2009.11.001

A18 Nicole Voges and Laurent U Perrinet. 'Phase space analysis of networks based on biologically realistic parameters'. In: *Journal of Physiology-Paris* 104.1-2 (10th Nov. 2010), pp. 51–60. ISSN: 1769-7115. DOI: 10.1016/j.jphysparis.2009.11.004. URL: http://dx.doi.org/10.1016/j.jphysparis.2009.11.004

A17 Jens Kremkow, Laurent U Perrinet, Guillaume S Masson and Ad M Aertsen. 'Functional consequences of correlated excitatory and inhibitory conductances in cortical networks'. In: *Journal of Computational Neuroscience* 28.3 (1st June 2010), pp. 579–94. DOI: 10.1007/s10827-010-0240-9. URL: http://www.ncbi.nlm.nih.gov/pubmed/20490645

A16 Khaled Masmoudi, Marc Antonini, Pierre Kornprobst, Laurent U Perrinet A novel bio-inspired static image compression scheme for noisy data transmission over low-bandwidth channels. *Acoustics Speech and Signal Processing (ICASSP)*, 2010.

2008

A15 Andrew P Davison, Daniel Bruderle, Jochen Eppler, Jens Kremkow, Eilif Muller, Dejan Pecevski, Laurent U Perrinet and Pierre Yger. 'PyNN: A Common Interface for Neuronal Network Simulators'. In: *Frontiers in Neuroinformatics* 2 (2008), p. 11. ISSN: 16625196. DOI: 10.3389/neuro.11.011.2008. URL: http://dx.doi.org/10.3389/neuro.11.011.2008

A14 Laurent U Perrinet. 'Adaptive Sparse Spike Coding : applications of Neuroscience to the compression of natural images'. In: *Optical and Digital Image Processing Conference 7000 - Proceedings of SPIE Volume 7000, 7 - 11 April 2008*. Ed. by Gabriel C. Peter Schelkens. Vol. 7000. 1. SPIE, 2008. DOI: 10.1117/12.787076

A13 Frédéric V Barthélemy, Laurent U Perrinet, Eric Castet and Guillaume S Masson. 'Dynamics of distributed 1D and 2D motion representations for short-latency ocular following'. In: *Vision research* 48.4 (1st Feb. 2008), pp. 501–522. ISSN: 0042-6989. DOI: 10.1016/j.visres.2007.10.020. URL: http://dx.doi.org/10.1016/j.visres.2007.10.020

2007

A12 Sylvain Fischer, Filip roubek, Laurent U Perrinet, Rafael Redondo and Gabriel Cristóbal. 'Self-Invertible 2D Log-Gabor Wavelets'. In: *International Journal of Computer Vision* 75.2 (13th Jan. 2007), pp. 231–246. ISSN: 1573-1405. DOI: 10.1007/s11263-006-0026-8. URL: http://dx.doi.org/10.1007/s11263-006-0026-8

A11 Sylvain Fischer, Rafael Redondo, Laurent U Perrinet and Gabriel Cristóbal. 'Sparse Approximation of Images Inspired from the Functional Architecture of the Primary Visual Areas'. In: *EURASIP Journal on Advances in Signal Processing* 2007.1 (2007), pp. 090727–122. ISSN: 1687-6180. DOI: 10.1155/2007/90727. URL: http://dx.doi.org/10.1155/2007/90727

A10 Anna Montagnini, Pascal Mamassian, Laurent U Perrinet, Eric Castet and Guillaume S Masson. 'Bayesian modeling of dynamic motion integration'. In: *Journal of Physiology-Paris* 101.1-3 (1st Jan. 2007), pp. 64–77. ISSN: 0928-4257. DOI: 10.1016/j.jphysparis.2007.10.013. URL: http://dx.doi.org/10.1016/j.jphysparis.2007.10.013

A9 Laurent U Perrinet and Guillaume S Masson. 'Modeling spatial integration in the ocular following response using a probabilistic framework'. In: *Journal of Physiology-Paris* 101.1–3 (2007), pp. 46–55. DOI: 10.1016/j.jphysparis.2007.10.011. URL: http://dx.doi.org/10.1016/j.jphysparis.2007.10.011

2004

A8 Laurent U Perrinet. 'Finding Independent Components using spikes : a natural result of Hebbian learning in a sparse spike coding scheme'. In: *Natural Computing* 3.2 (1st Jan. 2004), pp. 159–75. DOI: 10.1023/B:NACO.0000027753.27593.a7. URL: http://dx.doi.org/10.1023/B:NACO.0000027753.27593.a7

A7 Laurent U Perrinet. 'Feature detection using spikes : the greedy approach'. In: *Journal of Physiology-Paris* 98.4-6 (1st July 2004), pp. 530–9. DOI: 10.1016/j.jphysparis.2005.09.012. URL: http://dx.doi.org/10.1016/j.jphysparis.2005.09.012

A6 Laurent U Perrinet, Manuel Samuelides and Simon J Thorpe. 'Coding static natural images using spiking event times: do neurons cooperate?' In: *IEEE Transactions on Neural Networks* 15.5 (1st Sept. 2004). Special issue on 'Temporal Coding for Neural Information Processing', pp. 1164–75. DOI: 10.1109/TNN.2004.833303. URL: https://ieeexplore.ieee.org/document/1333080

2003

A5 Laurent U Perrinet, Manuel Samuelides and Simon J Thorpe. 'Emergence of filters from natural scenes in a sparse spike coding scheme'. In: *Neurocomputing* 58–60.C (2003). Special issue: Computational Neuroscience: Trends in Research 2004 - Edited by E. De Schutter, pp. 821–6. DOI: 10.1016/j.neucom.2004.01.133. URL: http://dx.doi.org/10.1016/j.neucom.2004.01.133

2002

A4 Laurent U Perrinet, Manuel Samuelides and Simon J Thorpe. 'Sparse spike coding in an asynchronous feed-forward multi-layer neural network using matching pursuit'. In: *Neurocomputing* 57 (1st Mar. 2004). Special issue: New Aspects in Neurocomputing: 10th European Symposium on Artificial Neural Networks 2002 - Edited by T. Villmann, pp. 125–134. ISSN: 0925-2312. DOI: 10.1016/j.neucom.2004.01.010. URL: http://dx.doi.org/10.1016/j.neucom.2004.01.010

A3 Laurent U Perrinet, Arnaud Delorme, Simon J Thorpe and Manuel Samuelides. 'Network of integrate-and-fire neurons using Rank Order Coding A: how to implement spike timing dependant plasticity'. In: *Neurocomputing* 38–40.1–4 (2001), pp. 817–22. DOI: 10.1016/S0925-2312(01)00460-X

A2 Arnaud Delorme, Laurent U Perrinet, Simon J Thorpe and Manuel Samuelides. 'Network of integrate-and-fire neurons using Rank Order Coding B: spike timing dependant plasticity and emergence of orientation selectivity'. In: *Neurocomputing* 38–40.1–4 (2001), pp. 539–45. DOI: 10.1016/s0925-2312(01)00403-9. URL: https://cerco.cnrs.fr/pagesp/arno/mypapers/Delorme.cns2000.pdf

A1 Laurent U Perrinet and Manuel Samuelides. 'Coherence detection in a spiking neuron via Hebbian learning'. In: *Neurocomputing* 44–46.C (1st June 2002), pp. 817–22. DOI: 10.1016/S0925-2312(02)00374-0. URL: http://dx.doi.org/10.1016/S0925-2312(02)00374-0

3 Peer-reviewed book chapters

C6 Laurent U Perrinet. 'From the retina to action: Dynamics of predictive processing in the visual system'. In: *The Philosophy and Science of Predictive Processing*. Ed. by Dina Mendonça, Manuel Curado and Steven S Gouveia. Bloomsbury, 2020, pp. 85–104. DOI: 10.5040/9781350099784.ch-005. URL: https://laurentperrinet.github.io/Perrinet20PredictiveProcessing_manubot/v/latest/index.html

C5 Anna Montagnini, Laurent U Perrinet and Guillaume S Masson. 'Visual motion processing and human tracking behavior'. In: *Biologically Inspired Computer Vision*. Ed. by Gabriel Cristóbal, Laurent U Perrinet and Matthias S Keil. Wiley-VCH Verlag GmbH and Co. KGaA, 1st Nov. 2015. Chap. 12. DOI: 10.1002/9783527680863.ch12. URL: https://laurentperrinet.github.io/publication/montagnini-15-bicv/

C4 Laurent U Perrinet. 'Sparse Models for Computer Vision'. In: *Biologically Inspired Computer Vision*. Ed. by Gabriel Cristóbal, Laurent U Perrinet and Matthias S Keil. Wiley-VCH Verlag GmbH and Co. KGaA, 1st Nov. 2015. Chap. 13. ISBN: 9783527680863. DOI: 10.1002/9783527680863.ch14. URL: http://onlinelibrary.wiley.com/doi/10.1002/9783527680863.ch14/summary

C3 Gabriel Cristóbal, Laurent U Perrinet and Matthias S Keil. 'Introduction'. In: *Biologically Inspired Computer Vision*. Ed. by Gabriel Cristóbal, Laurent U Perrinet and Matthias S Keil. Wiley-VCH Verlag GmbH and Co. KGaA, 1st Nov. 2015. Chap. 1. DOI: 10.1002/9783527680863.ch1. URL: http://bicv.github.io/chap1/

C2 Bruno Cessac, Emmanuel Daucé, Laurent U Perrinet and Manuel Samuelides. 'Introduction to Topics in Dynamical Neural Networks: From Large Scale Neural Networks to Motor Control and Vision'. In: *Topics in Dynamical Neural Networks: From Large Scale Neural Networks to Motor Control and Vision*. Vol. 142. The European Physical Journal Special Topics 1. Springer Verlag, 1st Mar. 2007, pp. 1–5. DOI: 10.1140/epjst/e2007-00057-3. URL: http://www.springerlink.com/index/10.1140/epjst/e2007-00057-3

C1 Laurent U Perrinet. 'Dynamical Neural Networks: modeling low-level vision at short latencies'. In: *Topics in Dynamical Neural Networks: From Large Scale Neural Networks to Motor Control and Vision*. Vol. 142. The European Physical Journal (Special Topics) 1. Berlin / Heidelberg: Springer Verlag, 1st Mar. 2007, pp. 163–225. DOI: 10.1140/epjst/e2007-00061-7

4 Theses, projects and books

B6 Laurent U Perrinet. 'La vision comme processus prédictif: Une approche biomimétique'. Programme de recherche (concours DR CNRS). Centre National de la Recherche Scientifique, France, 7th Jan. 2020. URL: https://laurentperrinet.github.io/publication/perrinet-20-dr/

B5 Gabriel Cristóbal, Laurent U Perrinet and Matthias S Keil, eds. *Biologically Inspired Computer Vision*. Weinheim, Germany: Wiley-VCH Verlag GmbH and Co. KGaA, 7th Oct. 2015. ISBN: 9783527680863. DOI: 10.1002/9783527680863. URL: http://onlinelibrary.wiley.com/book/10.1002/9783527680863

B4 Laurent U Perrinet. 'Codage prédictif dans les transformations visuo-motrices'. Habilitation à diriger des recherches (HDR). Aix-Marseille Université, Marseille, France, 17th Apr. 2014. URL: https://laurentperrinet.github.io/post/2014-04-17_hdr/

B3 Laurent U Perrinet and Emmanuel Daucé, eds. *Proceedings of the second french conference on Computational Neuroscience, Marseille*. 1st Oct. 2008. URL: https://hal.science/NEUROCOMP08

B2 Bruno Cessac, Emmanuel Daucé, Laurent U Perrinet and Manuel Samuelides. *Topics in Dynamical Neural Networks: From Large Scale Neural Networks to Motor Control and Vision*. Vol. 142. The European Physical Journal (Special Topics) 1. Berlin / Heidelberg: Springer Verlag, 1st Mar. 2007

B1 Laurent U Perrinet. 'Comment déchiffrer le code impulsionnel de la vision ? Étude du flux parallèle, asynchrone et épars dans le traitement visuel ultra-rapide'. PhD thesis. Université Paul Sabatier, Toulouse, France, 2003. URL: https://laurentperrinet.github.io/publication/perrinet-03-these

5 Supervised PhD theses

T9 Jean-Nicolas Jérémie. 'Foveal Retinotopy and Dual Pathways: A Computational Model for Active Visual Search'. Co-encadrement avec Emmanuel Daucé. thesis. Aix-Marseille Université, 10th Oct. 2025

T8 Antoine Grimaldi. 'Vision dynamique utilisant la précision temporelle des motifs d'impulsions dans les calculs neuronaux'. thesis. Aix-Marseille Université, 16th May 2024. URL: https://theses.fr/s380373

T7 Hugo Ladret. 'Modélisation multi-échelle de la sélectivité à l'orientation dans les stimulations visuelles naturelles'. thesis. Aix-Marseille Université, 8th Feb. 2024. URL: https://theses.fr/s377438

T6 Amélie Gruel. 'Réseaux de Neurones Impulsionnels Pour La Vision Embarquée Basée Sur Les Événements'. Co-encadrement avec Jean Martinet. These de doctorat. Université Côte d'Azur, 6th Oct. 2023. URL: https://theses.fr/2023COAZ4070

T5 Angelo Franciosini. 'SDPC : A Sparse and Predictive Model of the Early Visual System'. These de doctorat. Aix-Marseille Université, 28th Sept. 2021. URL: https://theses.fr/2021AIXM0346

T4 Kiana Mansour Pour. 'Effet de La Variabilité de La Vitesse Sur Le Mouvement de Poursuite Oculaire Lente et Sur La Perception de La Vitesse'. Co-encadrement avec Anna Montagnini. These de doctorat. Aix-Marseille Université, 1st Apr. 2019. DOI: 10.70675/d53c5b36zd868z43c3z889cz7924eb1417cf. URL: https://theses.fr/2019AIXM0137

T3 Jean-Bernard Damasse. 'Smooth Pursuit Eye Movements and Learning : Role of Motion Probability and Reinforcement Contingencies'. Co-encadrement avec Anna Montagnini. These de doctorat. Aix-Marseille Université, 11th June 2018. DOI: 10.70675/193e9600zb9f9z4bcazb4c1z4369b97dce0c. URL: https://theses.fr/2018AIXM0223

T2 Mina Aliakbari khoei. 'Une Approche Computationnelle de La Dépendance Au Mouvement Du Codage de La Position Dans La Système Visuel'. These de doctorat. Aix-Marseille Université, 6th Oct. 2014. URL: https://theses.fr/2014AIXM4041

T1 Jens Oliver Kremkow. 'Correlating Excitation and Inhibition in Visual Cortical Circuits : Functional Consequences and Biological Feasibility'. Co-encadrement avec Guillaume Masson. These de doctorat. Aix-Marseille 2, 1st Jan. 2009. URL: https://theses.fr/2009AIX20677

6 Peer-reviewed international conference proceedings

147. Alexis Gleyo and Laurent U Perrinet. 'Bio-plausible statistical regularizer for spiking LeJEPA'. in: *Flash talk at SNUFA 2026 online workshop (5-6 Nov 2026)*. Nov. 2026. URL: https://laurentperrinet.github.io/publication/gleyo-26-snufa/

146. Laurent U Perrinet. 'Working Memory in Recurrent Spiking Neural Networks With Heterogeneous Synaptic Delays'. In: *35th International Conference on Artificial Neural Networks (ICANN 2026) Special Session on Recent Advances in Spiking Neural Networks*. Padua (Italia), 14th Sept. 2026. DOI: 10.1007/978-3-032-38398-3_22. URL: https://laurentperrinet.github.io/publication/perrinet-26-icann/

145. Antoine Grimaldi, Matthieu Gilson, Laurent U Perrinet, Andrea Alamia, Boris Sotomayor-Gomez and Martin Vinck. 'Optimal Transport Theory to extract Spiking Motifs'. In: *Proceedings of the FENS Forum 2026*. S05-547. Poster Session 02 - Innovative Approaches in Computational Neuroscience. Barcelona (Spain), 7th July 2026. DOI: 10.21203/rs.3.rs-9518938/v1. URL: https://laurentperrinet.github.io/publication/grimaldi-26-fens/

144. Alexandre Lainé, Nicholas J. Priebe, Guillaume S. Masson and Laurent U Perrinet. 'Population decoding of visual motion direction'. In: *Proceedings of AREADNE*. Milos (Greece), 2026. URL: https://laurentperrinet.github.io/publication/laine-26-areadne

143. Laurent U Perrinet. 'Learning Working Memory in Recurrent Spiking Neural Networks Using Heterogeneous Delays'. In: *Proceedings of the Austrian Symposium on AI, Robotics and Vision*. Leoben (Austria), 12th Apr. 2026. URL: https://laurentperrinet.github.io/publication/perrinet-26-airov/

142. A Choplin, T Rakotomamonjy, Laurent U Perrinet, N Lantos and S Angelliaume. 'Physiological state Monitoring: a Riemannian Geometry based-model'. In: *The 14th International Winter Conference on Brain-Computer Interface, February 23 25, 2026*. High1 Resort (Korea), 2026. DOI: 10.1109/BCI69045.2026.11435104. URL: https://ieeexplore.ieee.org/document/11435104

141. Thomas Kronland-Martinet, Stéphane Viollet and Laurent U Perrinet. 'Detection of spiking motifs of arbitrary length in neural activity using bounded synaptic delays'. In: *Poster at SNUFA 2025 online workshop (5-6 Nov 2025)*. Nov. 2025. URL: https://snufa.net/2025/abstracts/thomas-kronland-martinet-detection.html

140. Alexandre Lainé, Sophie Denève, Nicholas J. Priebe, Guillaume S. Masson and Laurent U Perrinet. 'Population decoding of visual motion direction in V1 marmoset monkey : effects of uncertainty'. In: *Annual Computational Neuroscience Meeting: CNS*2025, Firenze, Italy*. 2025. URL: https://laurentperrinet.github.io/publication/laine-25-cns

139. A Choplin, T Rakotomamonjy, Laurent U Perrinet, N Lantos and S Angelliaume. 'Classification of Mental Workload Spatial Effects using Riemannian Manifold'. In: *Computational Cognitive Neuroscience Society Meeting (CCN) 2025*. Amsterdam (Netherlands), 2025. URL: https://laurentperrinet.github.io/publication/choplin-25-ccn

138. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Integrating the What and Where Visual Pathways to Improve CNN Categorisation'. In: *Computational Cognitive Neuroscience Society Meeting (CCN) 2025*. Amsterdam (Netherlands), 2025. URL: https://laurentperrinet.github.io/publication/jeremie-25-ccn

137. Antoine Grimaldi, Matthieu Gilson, Laurent U Perrinet, Andrea Alamia, Boris Sotomayor-Gomez and Martin Vinck. 'Robust Unsupervised Learning of Spike Patterns with Optimal Transport Theory'. In: *Computational and Systems Neuroscience (Cosyne) 2025*. Poster Presentation [2-076]. 28th Mar. 2025. URL: https://laurentperrinet.github.io/publication/grimaldi-25-cosyne/

136. Nikos Gekas, Andrew Isaac Meso, Jonathan Vacher, Laurent U Perrinet, Pascal Mamassian and Guillaume S Masson. 'An open-source vision-science tool for the auto-regressive generation of dynamic stochastic textures Motion Clouds'. In: *European Conference on Visual Perception*. 27th Aug. 2024. URL: https://laurentperrinet.github.io/publication/gekas-24-ecvp/

135. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Retinotopy in CNN's implements Efficient Visual Search'. In: *Computational Cognitive Neuroscience Society Meeting (CCN) 2024*. Boston (USA), 2024. URL: https://2024.ccneuro.org/poster/?id=293

134. Nelson Cortes, Lamyae Ikan, Hugo Ladret, Laurent U Perrinet and Christian Casanova. 'Diverse Neuronal Responses to Visual Precision in Cat Cortical Area 21a: Unraveling the Complexity of Orientation Processing'. In: *Proceedings of the FENS Forum 2024*. Vienna (Austria), 27th June 2024. URL: https://laurentperrinet.github.io/publication/cortes-24-fens/

133. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Retinotopy in CNN's implements Efficient Visual Search'. In: *Proceedings of the FENS Forum 2024*. Vienna (Austria), 27th June 2024. URL: https://2024.ccneuro.org/poster/?id=293

132. Adrien Fois and Laurent U Perrinet. 'Self-Supervised Learning of Spiking Motifs in Neurobiological Data'. In: *Proceedings of the FENS Forum 2024*. Vienna (Austria), 27th June 2024. URL: https://laurentperrinet.github.io/publication/fois-24-fens/

131. Laurent U Perrinet. 'Accurate Detection of Spiking Motifs in Neurobiological Data by Learning Heterogeneous Delays of a Spiking Neural Network'. In: *Proceedings of the FENS Forum 2024*. Vienna (Austria), 27th June 2024. URL: https://laurentperrinet.github.io/publication/perrinet-24-fens/

130. Laurent U Perrinet. 'Accurate Detection of Spiking Motifs by Learning Heterogeneous Delays of a Spiking Neural Network'. In: *32nd International Conference on Artificial Neural Networks (ICANN 2023)- Special Session on Recent Advances in Spiking Neural Networks*. Heraklion (Crete, Greece), 27th Sept. 2023. DOI: 10.1007/978-3-031-44207-0_31

129. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Retinotopy improves the categorisation and localisation of visual objects in CNNs'. In: *32nd International Conference on Artificial Neural Networks (ICANN 2023)*. Heraklion (Crete, Greece), 26th Sept. 2023. DOI: 10.1007/978-3-031-44207-0_52. URL: https://laurentperrinet.github.io/publication/jeremie-23-icann

128. Urbano Miguel Nunes, Laurent U Perrinet and Sio-Hoi Ieng. 'Time-to-Contact Map by Joint Estimation of Up-to-Scale Inverse Depth and Global Motion using a Single Event Camera'. In: *International Conference on Computer Vision 2023 (ICCV2023)*. 6th Oct. 2023. DOI: 10.1109/ICCV51070.2023.02162. URL: https://laurentperrinet.github.io/publication/nunes-23-iccv/

127. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Retinotopy improves the categorisation and localisation of visual objects in CNNs'. In: *Computational Cognitive Neuroscience Society Meeting (CCN) 2023*. Oxford (UK), 2023. DOI: 10.32470/ccn.2023.1706-0. URL: https://laurentperrinet.github.io/publication/jeremie-23-ccn

126. Hugo Ladret and Laurent U Perrinet. 'Convolutional Sparse Coding is improved by heterogeneous uncertainty modeling'. In: *ICLR 2023 SNN Workshop*. 5th May 2023. URL: https://laurentperrinet.github.io/publication/ladret-23-iclr/

125. Antoine Grimaldi and Laurent U Perrinet. 'Learning heterogeneous delays of spiking neurons for motion detection'. In: *GDR Vision, Toulouse, 2023*. Toulouse, France, 27th Jan. 2023. URL: https://gdr-vision-2023.sciencesconf.org/browse?forward-action=index&forward-controller=browse&docid=442297&lang=en

124. Hugo Ladret, Nelson Cortes, Lamyae Ikan, Frédéric Chavane, Christian Casanova and Laurent U Perrinet. 'Resilience to sensory uncertainty in the primary visual cortex'. In: *Computational and Systems Neuroscience (Cosyne) 2023*. 9th Mar. 2023. URL: https://www.world-wide.org/cosyne-23/resilience-sensory-uncertainty-primary-88600879/

123. Antoine Grimaldi, Camille Besnainou, Hugo Ladret and Laurent U Perrinet. 'Learning heterogeneous delays of spiking neurons for motion detection'. In: *Proceedings of ICIP 2022*. Bordeaux, France, 16th Oct. 2022. DOI: 10.1109/ICIP46576.2022.9897394. URL: https://ieeexplore.ieee.org/document/9897394/

122. Antoine Grimaldi and Laurent U Perrinet. 'Learning heterogeneous delays of spiking neurons for motion detection'. In: *Proceedings of the FENS Forum 2022*. S05-547. Poster Session 05 - Section: Modeling the Brain. Paris (France), 12th July 2022. URL: https://laurentperrinet.github.io/publication/grimaldi-22-fens/

121. Hugo Ladret and Laurent U Perrinet. 'Recurrent cortical connectivity in the primary visual cortex supports robust encoding of natural sensory inputs'. In: *Proceedings of the FENS Forum 2022*. S02-528. Poster Session 04 - Section: Visual System, From Retina to Cortex (ID 567). Paris (France), 11th July 2022. URL: https://laurentperrinet.github.io/publication/ladret-22-fens/

120. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Ultra-rapid visual search in natural images using active deep learning'. In: *Proceedings of the FENS Forum 2022*. S02-528. Poster Session 02 - Section: Machine Learning for Neuroscience and Psychiatry (ID 529). Paris (France), 10th July 2022. URL: https://laurentperrinet.github.io/publication/jeremie-22-fens/

119. Ghassan Dabane, Laurent U Perrinet and Emmanuel Daucé. 'What You See Is What You Transform: Foveated Spatial Transformers as a Bio-Inspired Attention Mechanism'. In: *IJCNN 2022 : International Joint Conference on Neural Networks*. 2022. DOI: 10.36227/techrxiv.16550391. URL: https://www.techrxiv.org/articles/preprint/What_You_See_Is_What_You_Transform_Foveated_Spatial_Transformers_as_a_bio-inspired_attention_mechanism/16550391/1

118. Hugo Ladret and Laurent U Perrinet. 'A resilient neural code in V1 to process natural images'. In: *Proceedings of AREADNE*. Santorini (Greece), 29th June 2022. URL: https://areadne.org/

117. Ilias Rentzeperis, Luca Calatroni, Laurent U Perrinet and Dario Prandi. 'Which sparsity problem does the brain solve?' In: *Proceedings of AREADNE*. Santorini (Greece), 29th June 2022. URL: https://laurentperrinet.github.io/publication/rentzeperis-22-areadne/

116. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Ultra-rapid visual search in natural images using active deep learning'. In: *Proceedings of AREADNE*. Santorini (Greece), 29th June 2022. URL: https://areadne.org/

115. Antoine Grimaldi, Camille Besnainou, Hugo Ladret and Laurent U Perrinet. 'Decoding spiking motifs using neurons with heterogeneous delays'. In: *Proceedings of AREADNE*. Santorini (Greece), 29th June 2022. URL: https://areadne.org/

114. Hugo Ladret, Nelson Cortes, Lamyae Ikan, Frédéric Y Chavane, Christian Casanova and Laurent U Perrinet. 'Modulation of orientation selectivity by orientation precision'. In: *Proceedings of the Society for Neuroscience conference*. P465.08. 2021. URL: https://www.abstractsonline.com/pp8/#!/10485/presentation/22078

113. Hugo Ladret and Laurent U Perrinet. 'Decoding orientation distributions from noisy observations in V1'. In: *Champalimaud Research Symposium (CRS21)*. Poster Session II -Fri 15 Oct 13:30—15:30h, Lisbon time. 15th Oct. 2021

112. Jean-Nicolas Jérémie and Laurent U Perrinet. 'Ultra-fast categorization of images containing animals in vivo and in computo'. In: *Champalimaud Research Symposium (CRS21)*. Poster Session II -Fri 15 Oct 13:30—15:30h, Lisbon time. 15th Oct. 2021

111. Antoine Grimaldi, Victor Boutin, Sio-Hoi Ieng, Ryad Benosman and Laurent U Perrinet. 'From event-based computations to a bio-plausible Spiking Neural Network'. In: *Champalimaud Research Symposium (CRS21)*. Poster Session I Thu 14 Oct 14–16h, Lisbon time. 14th Oct. 2021

110. Antoine Grimaldi, Victor Boutin, Sio-Hoi Ieng, Laurent U Perrinet and Ryad Benosman. 'A robust bio-inspired approach to event-driven object recognition'. In: *Computational and Systems Neuroscience (Cosyne) 2021*. 26th Feb. 2021. URL: https://laurentperrinet.github.io/publication/grimaldi-21-cosyne/

109. Antoine Grimaldi, Victor Boutin, Sio-Hoi Ieng, Laurent U Perrinet and Ryad Benosman. 'A homeostatic gain control mechanism to improve event-driven object recognition'. In: *Content-Based Multimedia Indexing (CBMI) 2021*. 24th June 2021. DOI: 10.1109/CBMI50038.2021.9461901. URL: https://laurentperrinet.github.io/publication/grimaldi-21-cbmi/

108. Alberto Vergani and Laurent U Perrinet. 'Simulating anticipatory activity in a 1D Spiking Neural Network Model'. In: *Bernstein Conference 2021*. 22nd Sept. 2021. DOI: 10.12751/nncn.bc2021.p094

107. Hugo Ladret and Laurent U Perrinet. 'Learning dynamics in a neural network model of the primary visual cortex'. In: vol. 4. 0. 2020. URL: http://aes.amegroups.com/article/view/5214

106. Angelo Franciosini, Victor Boutin and Laurent U Perrinet. 'Modelling Complex-cells and topological structure in the visual cortex of mammals using Sparse Predictive Coding'. In: *SIGMA'2020 (Signal, Image, Geometry, Modelling, Approximation)*. 30th Mar. 2020. URL: https://conferences.cirm-math.fr/2152.html

105. Angelo Franciosini, Victor Boutin and Laurent U Perrinet. 'Modelling Complex-cells and topological structure in the visual cortex of mammals using Sparse Predictive Coding'. In: *Computational and Systems Neuroscience (Cosyne) 2020*. 27th Sept. 2020. URL: https://laurentperrinet.github.io/publication/franciosini-20-cosyne/

104. Emmanuel Daucé and Laurent U Perrinet. 'Visual search as active inference'. In: *IWAI 2020*. 17th Dec. 2020. DOI: 10.1007/978-3-030-64919-7_17. URL: https://laurentperrinet.github.io/publication/dauce-20-iwai

103. Victor Boutin, Angelo Franciosini, Frédéric Y Chavane, Franck Ruffier and Laurent U Perrinet. 'Sparse Deep Predictive Coding captures contour integration capabilities of the early visual system'. In: *SIGMA'2020 (Signal, Image, Geometry, Modelling, Approximation)*. 3rd Mar. 2019. URL: https://conferences.cirm-math.fr/2152.html

102. Hugo Ladret, Nelson Cortes, Frédéric Y Chavane, Laurent U Perrinet and Christian Casanova. 'Orientation selectivity to synthetic natural patterns in a cortical-like

model of the cat primary visual cortex'. In: *Proceedings of the Society for Neuroscience conference*. 403.16 / P20. 2019. URL: https://www.abstractsonline.com/pp8/#!/7883/presentation/65859

101. Wahiba Taouali, Giacomo Benvenuti, Frédéric Y Chavane and Laurent U Perrinet. 'A dynamic model for decoding direction and orientation in macaque primary visual cortex'. In: *Proceedings of NCCD, Capbreton*. 23rd Sept. 2019. URL: https://laurentperrinet.github.io/publication/perrinet-19-nccd

100. Victor Boutin, Angelo Franciosini, Frédéric Y Chavane, Franck Ruffier and Laurent U Perrinet. 'Sparse Deep Predictive Coding to model visual object recognition'. In: *Proceedings of the Society for Neuroscience conference*. presentation number : 490.02. 2019. URL: https://laurentperrinet.github.io/publication/boutin-franciosini-ruffier-perrinet-19-sfn/

99. Angelo Franciosini, Victor Boutin and Laurent U Perrinet. 'Modelling Complex Cells of Early Visual Cortex using Predictive Coding'. In: *Annual Computational Neuroscience Meeting: CNS*2019, Barcelona*. P243. 2019. URL: https://www.cnsorg.org/cns-2019-poster-presentation-guide

98. Angelo Franciosini, Victor Boutin and Laurent U Perrinet. 'A hierarchical, multilayer convolutional sparse coding algorithm based on predictive coding'. In: *NeuroFrance 2019, International Conference from the Société des Neurosciences, Marseille, France*. 2019. URL: https://www.professionalabstracts.com/nf2019/iplanner/#/presentation/790

97. Victor Boutin, Angelo Franciosini, Franck Ruffier and Laurent U Perrinet. 'Top-down connection in Hierarchical Sparse Coding'. In: *GdR Robotics - 2019-06-05*. 2019

96. Victor Boutin, Angelo Franciosini, Franck Ruffier and Laurent U Perrinet. 'Unsupervised Hierarchical Sparse Coding algorithm inspired by Biological Vision'. In: *Doc2AMU Doctoral Day - 2018-11-23*. 2018

95. Victor Boutin, Angelo Franciosini, Franck Ruffier and Laurent U Perrinet. 'From biological vision to unsupervised hierarchical sparse coding'. In: *iTwist, 2018*. 2018. URL: https://laurentperrinet.github.io/publication/boutin-franciosini-ruffier-perrinet-18-itwist/

94. Julien Dupeyroux, Victor Boutin, Julien R Serres, Laurent U Perrinet and Stéphane Viollet. 'M2APix: a bio-inspired auto-adaptive visual sensor for robust ground height estimation'. In: *ISCAS2018, IEEE International Symposium on Circuits and Systems*. 2018. DOI: 10.1109/iscas.2018.8351433. URL: https://ieeexplore.ieee.org/abstract/document/8351433

93. Angelo Franciosini and Laurent U Perrinet. 'On the Origins of Hierarchy in Visual Processing'. In: *Curves and Surfaces 2018, Arcachon*. 2018. URL: https://laurentperrinet.github.io/publication/franciosini-perrinet-18-cs/

92. Hugo Ladret and Laurent U Perrinet. 'Selectivity to oriented patterns of different precisions'. In: *GDR Vision, Paris, 2018*. 2018. URL: https://github.com/hugoladret/InternshipM1/raw/master/2018-06_POSTER_final.pdf

91. Kiana Mansour Pour, Nikos Gekas, Pascal Mamassian, Laurent U Perrinet, Anna Montagnini and Guillaume S Masson. 'Speed uncertainty and motion perception with naturalistic random textures'. In: *Journal of Vision, Vol.18, 345, proceedings of VSS*. 26.472. 2018. DOI: 10.1167/18.10.345. URL: https://laurentperrinet.github.io/publication/mansour-18-vss

90. Chloé Pasturel, Anna Montagnini and Laurent U Perrinet. 'Estimating and anticipating a dynamic probabilistic bias in visual motion direction'. In: 2018. URL: https://laurentperrinet.github.io/publication/pasturel-18

89. Chloé Pasturel, Anna Montagnini and Laurent U Perrinet. 'ANEMO: Quantitative tools for the ANalysis of Eye MOvements'. In: *Grenoble Workshop on Models and Analysis of Eye Movements, Grenoble, France*. 2018. URL: https://laurentperrinet.github.io/publication/pasturel-18-anemo

88. Laurent U Perrinet, Chloé Pasturel and Anna Montagnini. 'Estimating and anticipating a dynamic probabilistic bias in visual motion direction'. In: *Grenoble Workshop on Models and Analysis of Eye Movements, Grenoble, France*. 2018. URL: https://laurentperrinet.github.io/publication/pasturel-18-grenoble

87. Laurent U Perrinet. 'A low-cost, accessible eye tracking framework'. In: *GDR Vision, Paris, 2018*. 2018. URL: https://github.com/laurentperrinet/Perrinet18gdr

86. Victor Boutin, Angelo Franciosini, Franck Ruffier and Laurent U Perrinet. 'Controlling an aerial robot with human gestures using bio-inspired algorithm'. In: *Doc2AMU Doctoral Day - 2017-10-13*. 2017

85. Victor Boutin, Franck Ruffier and Laurent U Perrinet. 'Efficient learning of sparse image representations using homeostatic regulation'. In: *NeuroFrance 2017, International Conference from the Société des Neurosciences, Bordeaux, France*. 2017

84. Victor Boutin, Franck Ruffier and Laurent U Perrinet. 'Efficient learning of sparse image representations using homeostatic regulation'. In: *SPARS2017, Lisbon*. 2017

83. Kiana Mansour Pour, Laurent U Perrinet, Guillaume S Masson and Anna Montagnini. 'How the dynamics of human smooth pursuit is influenced by speed uncertainty'. In: *European Conference on Visual Perception*. 2017. URL: https://laurentperrinet.github.io/publication/mansour-17-ecvp/

82. Kiana Mansour Pour, Laurent U Perrinet, Guillaume S Masson and Anna Montagnini. 'Voluntary tracking the moving clouds : Effects of speed variability on human smooth pursuit'. In: *GDR Vision, Lille, 2017*. 2017. URL: https://laurentperrinet.github.io/publication/mansour-17-gdr

81. Chloé Pasturel, Jean-Bernard Damasse, Anna Montagnini and Laurent U Perrinet. 'Estimating and anticipating a dynamic probabilistic bias in visual motion direction'. In: *GDR Vision, Lille, 2017*. 2017. URL: https://laurentperrinet.github.io/publication/pasturel-17-gdr

80. Laurent U Perrinet and Etienne Rey. 'Expériences autour de la perception de la forme en art et science'. In: *GDR Vision, Lille, 2017*. 2017. URL: https://github.com/NaturalPatterns/2017-10-12_GDR

79. Jean-Bernard Damasse, Anna Montagnini and Laurent U Perrinet. 'Dynamic modulation of volatility by reward contingencies: effects on anticipatory smooth eye movement'. In: *Proceedings of Vision Sciences Society Annual Meeting*. Vol. 17. 12. Meeting abstract presented at VSS 2017. The Association for Research in Vision and Ophthalmology, 2017, p. 273. DOI: 10.1167/17.10.273. URL: http://jov.arvojournals.org/article.aspx?doi=10.1167/17.10.273

78. Laurent U Perrinet. 'Biologically-inspired characterization of sparseness in natural images'. In: *2016 6th European Workshop on Visual Information Processing (EUVIP)*. IEEE, 1st Oct. 2016, pp. 1–6. ISBN: 978-1-5090-2781-1. DOI: 10.1109/EUVIP.2016.7764592. URL: http://ieeexplore.ieee.org/document/7764592/

77. Kiana Mansour Pour, Laurent U Perrinet, Guillaume S Masson and Anna Montagnini. 'Voluntary tracking the moving clouds : Effects of speed variability on human smooth pursuit'. In: *Proceedings of the Society for Neuroscience conference*. 2016, 2P045. URL: https://laurentperrinet.github.io/publication/mansour-16-sfn

76. Kiana Mansour Pour, Laurent U Perrinet, Guillaume S Masson and Anna Montagnini. 'Voluntary tracking the moving clouds : Effects of speed variability on human smooth pursuit'. In: *GDR Vision, Toulouse, Nov 3rd, 2016*. 2016. URL: https://laurentperrinet.github.io/publication/mansour-16-gdr

75. Jean-Bernard Damasse, Anna Montagnini and Laurent U Perrinet. 'Modeling the effect of dynamic contingencies on anticipatory eye movements'. In: *European Conference on Visual Perception*. 2016, 2P044. URL: https://laurentperrinet.github.io/publication/damasse-16-ecvp

74. Kiana Mansour Pour, Laurent U Perrinet, Guillaume S Masson and Anna Montagnini. 'Voluntary tracking the moving clouds : Effects of speed variability on human smooth pursuit'. In: *European Conference on Visual Perception*. 2016, 2P045. URL: https://laurentperrinet.github.io/publication/mansour-16-ecvp

73. Jean-Bernard Damasse, Laurent U Perrinet, Jérémie Jozefowiez, Laurent Madelain and Anna Montagnini. 'Operant reinforcement versus reward expectancy: effects on anticipatory eye movements'. In: *Proceedings of VSS*. vol. 16. 12. The Association for Research in Vision and Ophthalmology, 1st Sept. 2016, p. 1356. DOI: 10.1167/16.12.1356. URL: http://jov.arvojournals.org/article.aspx?doi=10.1167/16.12.1356

72. Anna Montagnini, Jean-Bernard Damasse, Laurent U Perrinet and Guillaume S Masson. 'Effects of motion predictability on anticipatory and visually-guided eye movements: a common prior for sensory processing and motor control?' In: *European Conference on Visual Perception*. 2016, 22T106. URL: https://laurentperrinet.github.io/publication/montagnini-16-ecvp

71. Laurent U Perrinet, Rick A Adams and Karl Friston. 'Compensation of oculomotor delays in the visual system's network'. In: *Complex Networks: from theory to interdisciplinary applications*. 2016, paper 61. URL: https://laurentperrinet.github.io/publication/perrinet-16-networks

70. Wahiba Taouali, Giacomo Benvenuti, Frédéric Y Chavane and Laurent U Perrinet. 'A dynamic model for decoding direction and orientation in macaque primary visual cortex'. In: *Proceedings of AREADNE*. 2016. DOI: 10.1167/15.12.484. URL: http://jov.arvojournals.org/article.aspx?articleid=2433592

69. Cesar U Ravello, F. Olivares, R. Herzog, Laurent U Perrinet, Maria-José Escobar and Adrián G Palacios. 'Spatiotemporal tuning of retinal ganglion cells dependent on the context of signal presentation'. In: *European Retina Meeting 2015*. 2015

68. Jonathan Vacher, Andrew Isaac Meso, Laurent U Perrinet and Gabriel Peyré. 'A Mathematical Account of Dynamic Texture Synthesis for Probing Visual Perception'. In: *ICMS 2015 conference*. 2015

67. Laurent U Perrinet and James A Bednar. 'Sparse Coding Of Natural Images Using A Prior On Edge Co-Occurences'. In: *European Signal Processing Conference 2015 (EUSIPCO 2015)*. Nice, France, 1st Aug. 2015. DOI: 10.1109/EUSIPCO.2015.7362781. URL: http://dx.doi.org/10.1109/EUSIPCO.2015.7362781

66. Anna Montagnini, Jean-Bernard Damasse, Laurent U Perrinet and Laurent Madelain. 'Anticipating a moving target: role of vision and reinforcement'. In: *Proceedings of the Society for Neuroscience conference*. 2015. URL: https://laurentperrinet.github.io/publication/montagnini-15-sfn

65. Wahiba Taouali, Giacomo Benvenuti, Frédéric Y Chavane and Laurent U Perrinet. 'A dynamic model for decoding direction and orientation in macaque primary visual cortex'. In: *Proceedings of VSS*. 2016. DOI: 10.1167/15.12.484. URL: http://jov.arvojournals.org/article.aspx?articleid=2433592

64. Jean-Bernard Damasse, Laurent Madelain, Laurent U Perrinet and Anna Montagnini. 'Anticipatory smooth eye movements and reinforcement'. In: *Proceedings of VSS*. The Association for Research in Vision and Ophthalmology, 1st Sept. 2015. DOI: 10.1167/15.12.1019. URL: http://jov.arvojournals.org/article.aspx?articleid=2434129

63. Fréderic Danion, Caroline Landelle, Anna Montagnini, Laurent U Perrinet and Laurent Madelain. 'Eye tracking a self-moved target with complex hand-target dynamics'. In: *Proceedings of the Society for Neuroscience conference*. SfN. 2015. URL: https://laurentperrinet.github.io/publication/danion-15-sfn

62. Wahiba Taouali, Giacomo Benvenuti, Pascal Wallisch, Frédéric Y Chavane and Laurent U Perrinet. 'On overdispersion in neuronal evoked activity'. In: *ICMNS 2015 conference*. 2015

61. Jonathan Vacher, Andrew Isaac Meso, Laurent U Perrinet and Gabriel Peyré. 'Dynamic Textures For Probing Motion Perception'. In: *IHP workshop*. 2014

60. P Philipp Rudiger, Jean-Luc Stevens, Bharath Chandra Talluri, Laurent U Perrinet and James A Bednar. 'Relationship between natural image statistics and lateral connectivity in the primary visual cortex'. In: *Proceedings of COSYNE*. 2014. URL: http://goo.gl/RJpJR4

59. Laurent U Perrinet and James A Bednar. 'Edge co-occurrences are sufficient to categorize natural versus animal images'. In: vol. 14. 10. Association for Research in Vision and Ophthalmology, 22nd Aug. 2014, p. 1310. DOI: 10.1167/14.10.1310. URL: http://dx.doi.org/10.1167/14.10.1310

58. Claudio Simoncini, Anna Montagnini, Laurent U Perrinet and Guillaume S Masson. 'The characteristics of microsaccadic eye movements varied with the change of strategy in a match-to-sample task'. In: vol. 14. 10. Association for Research in Vision and Ophthalmology, 22nd Aug. 2014, p. 110. DOI: 10.1167/14.10.110. URL: http://dx.doi.org/10.1167/14.10.110

57. Bernhard A Kaplan, Mina A Khoei, Anders Lansner and Laurent U Perrinet. 'Signature of an anticipatory response in area V1 as modeled by a probabilistic model and a spiking neural network'. In: *IEEE International Joint Conference on Neural Networks (IJCNN) 2014 Beijing, China*. Bernhard A Kaplan and Mina A Khoei contributed equally to this work. 6th July 2014. DOI: 10.1109/IJCNN.2014.6889847. URL: https://laurentperrinet.github.io/publication/kaplan-khoei-14

56. Mina A Khoei, Laurent U Perrinet and Guillaume S Masson. 'Motion-based prediction model for flash lag effect'. In: vol. 14. 10. Association for Research in Vision and Ophthalmology, 22nd Aug. 2014, p. 471. DOI: 10.1167/14.10.471. URL: http://dx.doi.org/10.1167/14.10.471

55. Wahiba Taouali and Laurent U Perrinet. 'A Simple Model of Orientation Encoding Accounting For Multivariate Neural Noise'. In: *6th Workshop of the Computational Neuroscience Network in Marseille*. 2014

54. Wahiba Taouali and Laurent U Perrinet. 'A Simple Model of Orientation Encoding Accounting For Multivariate Neural Noise'. In: *Proceedings of AREADNE*. 2014

53. Andrew Isaac Meso, Claudio Simoncini, Laurent U Perrinet and Guillaume S Masson. 'Beyond simply faster and slower: exploring paradoxes in speed perception'. In: vol. 14. 10. Association for Research in Vision and Ophthalmology, 22nd Aug. 2014, p. 491. DOI: 10.1167/14.10.491. URL: http://dx.doi.org/10.1167/14.10.491

52. Mina A Khoei, Giacomo Benvenuti, Frédéric Y Chavane and Laurent U Perrinet. 'Motion-based prediction and development of the response to an 'on the way' stimulus'. In: *Annual Computational Neuroscience Meeting: CNS*2013, Paris*. 2013. DOI:

10.1186/1471-2202-14-S1-P314. URL: https://laurentperrinet.github.io/publication/khoei-13-cns

51. Laurent U Perrinet, Rick A Adams and Karl Friston. 'Active inference, eye movements and oculomotor delays'. In: *Annual Computational Neuroscience Meeting: CNS 2013, Paris*. 2013. URL: https://laurentperrinet.github.io/publication/perrinet-13-cns

50. Andrew Isaac Meso, Claudio Simoncini, Laurent U Perrinet and Guillaume S Masson. 'How and why do image frequency properties influence perceived speed?' In: *VSS Conference Abstract*. Vol. (13)9. 2013, p. 354. DOI: 10.1167/13.9.354. URL: https://laurentperrinet.github.io/publication/meso-13-vss

49. Laurent U Perrinet, Rick A Adams and Karl Friston. 'Active inference, eye movements and oculomotor delays'. In: *The 7th Japanese-French Frontiers of Science Symposium*. 2013. URL: https://laurentperrinet.github.io/publication/perrinet-13-jffos

48. Claudio Simoncini, Laurent U Perrinet, Anna Montagnini and Guillaume S Masson. 'Measuring speed of moving textures: Different pooling of motion information for human ocular following and perception'. In: *VSS Conference Abstract*. 2013. DOI: 10.3389/conf.fnins.2012.86.00016

47. Laurent U Perrinet, Rick A Adams and Karl Friston. 'Active inference, smooth pursuit and oculomotor delays'. In: *Proceedings of AREADNE, Santorini, Greece, 21-24 June 2012, published by The AREADNE Foundation, Inc., Cambridge, Massachusetts, USA, http://areadne.org*. 2012

46. Guillaume S Masson and Laurent U Perrinet. 'Motion-based prediction is sufficient to solve the aperture problem'. In: *Proceedings of AREADNE*. 2012. URL: https://laurentperrinet.github.io/publication/masson-12-areadne

45. Mina A Khoei, Laurent U Perrinet and Guillaume S Masson. 'Role of motion-based prediction in motion extrapolation'. In: *Proceedings of the Society for Neuroscience conference*. SfN. 2012. URL: https://laurentperrinet.github.io/publication/khoei-12-sfn

44. Claudio Simoncini, Laurent U Perrinet, Anna Montagnini, Pascal Mamassian and Guillaume S Masson. 'Measuring speed of moving textures: Different pooling of motion information for human ocular following and perception.' In: *Front. Neurosci. Conference Abstract: Neural Coding, Decision-Making and Integration in Time*. 2012. DOI: 10.3389/conf.fnins.2012.86.00016. URL: http://www.frontiersin.org/myfrontiers/abstractdetails.aspx?abs_doi=10.3389/conf.fnins.2012.86.00016

43. Claudio Simoncini, Anna Montagnini, Laurent U Perrinet and Guillaume S Masson. 'Effect of image statistics on fixational eye movements'. In: *VSS Conference Abstract*. 2012. DOI: 10.1167/12.9.1014. URL: http://www.journalofvision.org/content/12/9/1014.abstract?sid=9c51ff88-5b9a-4d1b-aaf1-a1219bd02b0a

42. Claudio Simoncini, Anna Montagnini, Laurent U Perrinet, Pascal Mamassian and Guillaume S Masson. 'Pattern discrimination for moving random textures: Richer stimuli are more difficult to recognize'. In: vol. 11. 11. Association for Research in Vision and Ophthalmology, 23rd Sept. 2011, p. 749. DOI: 10.1167/11.11.749. URL: http://dx.doi.org/10.1167/11.11.749

41. Claudio Simoncini, Anna Montagnini, Laurent U Perrinet, Pascal Mamassian and Guillaume S Masson. 'Pattern discrimination for moving random textures: Richer stimuli are more difficult to recognize'. In: *VSS Conference Abstract*. 1st Aug. 2012. DOI: 10.1167/11.11.749. URL: http://www.journalofvision.org/content/12/9/1014.abstract?sid=9c51ff88-5b9a-4d1b-aaf1-a1219bd02b0a

40. Laurent U Perrinet, David Fitzpatrick and James A Bednar. 'Edge statistics in natural images versus laboratory animal environments: implications for understanding lateral connectivity in V1'. In: *Proceedings of the Society for Neuroscience conference*. Ed. by Www Washington. Program No. 530.04. 2011. URL: https://laurentperrinet.github.io/publication/perrinet-11-sfn

39. Claudio Simoncini, Laurent U Perrinet, Anna Montagnini, Pascal Mamassian and Guillaume S Masson. 'Different pooling of motion information for perceptual speed discrimination and behavioral speed estimation'. In: *Vision Science Society*. 43.503. 2010. DOI: 10.1167/10.7.834

38. Laurent U Perrinet. 'Probabilistic models of the low-level visual system: the role of prediction in detecting motion'. In: *LADISLAV TAUC and GDR MSPC NEUROSCIENCES CONFERENCE, From Mathematical Image Analysis to Neurogeometry of the Brain*. 2010. URL: https://laurentperrinet.github.io/publication/perrinet-10-tauc/

37. Laurent U Perrinet and Guillaume S Masson. 'Dynamical emergence of a neural solution for motion integration'. In: *Proceedings of AREADNE*. 2010

36. Amarender Bogadhi, Anna Montagnini, Pascal Mamassian, Laurent U Perrinet and Guillaume S Masson. 'A recurrent Bayesian model of dynamic motion integration for smooth pursuit'. In: *Vision Science Society*. 26.445. 2010. DOI: 10.1167/10.7.545. URL: http://dx.doi.org/10.1167/10.7.545

35. Nicole Voges and Laurent U Perrinet. 'Phase space analysis of networks based on biologically realistic parameters'. In: *Proceedings of NeuroComp*. Vol. 104. 1-2. 2010, pp. 51–60. DOI: 10.1016/j.jphysparis.2009.11.004

34. Laurent U Perrinet, Alexandre Reynaud, Frédéric Y Chavane and Guillaume S Masson. 'Inferring monkey ocular following responses from V1 population dynamics using a probabilistic model of motion integration'. In: *Vision Science Society*. 23.411. 2009. DOI: 10.1167/9.8.745

33. Laurent U Perrinet, Nicole Voges, Jens Kremkow and Guillaume S Masson. 'Decoding center-surround interactions in population of neurons for the ocular following response'. In: *Proceedings of COSYNE*. 2009. DOI: 10.3389/conf.neuro.06.2009.03.266

32. Nicole Voges and Laurent U Perrinet. 'Dynamical state spaces of cortical networks representing various horizontal connectivities'. In: *Proceedings of COSYNE*. 2009. DOI: 10.3389/conf.neuro.06.2009.03.207

31. Nicole Voges and Laurent U Perrinet. 'Dynamics of cortical networks including long-range patchy connections'. In: *Eighth Göttingen Meeting of the German Neuroscience Society*. 2009, T26–3C

30. Jens Kremkow, Laurent U Perrinet, Guillaume S Masson and Ad M Aertsen. 'Functional consequences of correlated excitation and inhibition on single neuron integration and signal propagation through synfire chains'. In: *Eighth Göttingen Meeting of the German Neuroscience Society*. 2009, T26–6B

29. Pierre Yger, Daniel Bruderle, Jochen Eppler, Jens Kremkow, Dejan Pecevski, Laurent U Perrinet, Michael Schmuker, Eilif Muller and Andrew P Davison. 'NeuralEnsemble: Towards a meta-environment for network modeling and data analysis'. In: *Eighth Göttingen Meeting of the German Neuroscience Society*. 2009, T26–4C

28. Jens Kremkow, Laurent U Perrinet, Pierre Baudot, Manu Levy, Olivier Marre, Cyril Monier, Yves Frégnac, Guillaume S Masson and Ad M Aertsen. 'Control of the temporal interplay between excitation and inhibition by the statistics of visual input: a

V1 network modelling study'. In: *Proceedings of the Society for Neuroscience conference*. 2008

27. Nicole Voges and Laurent U Perrinet. 'Analyzing cortical network dynamics with respect to different connectivity assumptions'. In: *Proceedings of NeuroComp08, Marseille*. Ed. by Laurent U Perrinet and Emmanuel Daucé. 1st Oct. 2008

26. Nicole Voges, Jens Kremkow and Laurent U Perrinet. 'Dynamics of cortical networks based on patchy connectivity patterns'. In: *Proceedings of the FENS Forum 2008*. Vol. 4. 075.14. 2008

25. Jens Kremkow, Laurent U Perrinet, Ad M Aertsen and Guillaume S Masson. 'Functional properties of feed-forward inhibition'. In: *Proceedings of NeuroComp08, Marseille*. Ed. by Laurent U Perrinet and Emmanuel Daucé. 1st Oct. 2008

24. Laurent U Perrinet and Guillaume S Masson. 'Modeling spatial integration in the ocular following response to center-surround stimulation using a probabilistic framework'. In: *Proceedings of COSYNE*. 2008. DOI: 10.1016/j.jphysparis.2007.10.011

23. Laurent U Perrinet and Guillaume S Masson. 'Decoding the population dynamics underlying ocular following response using a probabilistic framework'. In: *Proceedings of AREADNE, 2008*. 2008

22. Laurent U Perrinet. 'What adaptive code for efficient spiking representations? A model for the formation of receptive fields of simple cells'. In: *Proceedings of COSYNE*. 2008

21. Andrew P Davison, Pierre Yger, Jens Kremkow, Laurent U Perrinet and Eilif Muller. 'PyNN: towards a universal neural simulator API in Python'. In: *Sixteenth Annual Computational Neuroscience Meeting: CNS*2007, Toronto, Canada. 7–12 July 2007*. Ed. by B. M. C. Neuroscience. Vol. 8(Suppl 2):P2. 2007. DOI: 10.1186/1471-2202-8-S2-P2. URL: http://dx.doi.org/10.1186/1471-2202-8-S2-P2

20. Jens Kremkow, Laurent U Perrinet, Arvind Kumar, Ad M Aertsen and Guillaume S Masson. 'Synchrony in thalamic inputs enhances propagation of activity through cortical layers'. In: *Annual Computational Neuroscience Meeting: BMC Neuroscience*. Ed. by BMC Neuroscience. Vol. 8. Suppl 2. 6th July 2007, P180+. DOI: 10.1186/1471-2202-8-S2-P180. URL: http://dx.doi.org/10.1186/1471-2202-8-S2-P180

19. Anna Montagnini, Pascal Mamassian, Laurent U Perrinet, Eric Castet and Guillaume S Masson. 'Dynamic inference for motion tracking'. In: *European Conference on Visual Perception*. 2007

18. Anna Montagnini, Pascal Mamassian, Laurent U Perrinet and Guillaume S Masson. 'Visual tracking of ambiguous moving objects: A recursive Bayesian model'. In: *Journal of Vision*. Vol. 7. 9. 2007, p. 406. URL: https://jov.arvojournals.org/article.aspx?articleid=2134393

17. Laurent U Perrinet. 'On efficient sparse spike coding schemes for learning natural scenes in the primary visual cortex'. In: *Sixteenth Annual Computational Neuroscience Meeting: CNS*2007, Toronto, Canada. 7–12 July 2007*. Ed. by Bmc N. 2007. Vol. 8(Suppl 2):P206. This work is supported by the 6th RFP of the EU (grant no. 15879-FACETS).. 2007. DOI: 10.1186/1471-2202-8-S2-P206. URL: http://dx.doi.org/10.1186/1471-2202-8-S2-P206

16. Laurent U Perrinet. 'Neural Codes for Adaptive Sparse Representations of Natural Images'. In: *Mathematical image processing meeting (Marseille, France) September 5, 2007*. 2007

15. Laurent U Perrinet and Jens Kremkow. 'Dynamical contrast gain control mechanisms in a layer 2/3 model of the primary visual cortex'. In: *Physiogenic and pathogenic oscillations: the beauty and the beast, 5th INMED/TINS CONFERENCE SEPTEMBER 9 - 12, 2006, La Ciotat, France*. 2006

14. Laurent U Perrinet. 'An efficiency razor for model selection and adaptation in the primary visual cortex'. In: *Fifteenth Annual Computational Neuroscience Meeting: CNS*2006*. 2006. URL: https://ocns.memberclicks.net/assets/docs/CNS_Program_books/2006booklet.pdf

13. Laurent U Perrinet, Frédéric V Barthélemy and Guillaume S Masson. 'Input-output transformation in the visuo-oculomotor loop: modeling the ocular following response to center-surround stimulation in a probabilistic framework'. In: *1ère conférence francophone NEUROsciences COMPutationnelles - NeuroComp*. 2006

12. Laurent U Perrinet and Jens Kremkow. 'Dynamical contrast gain control mechanisms in a layer 2/3 model of the primary visual cortex'. In: *The Functional Architecture of the Brain : from Dendrites to Networks. Symposium in honour of Dr Suzanne Tyc-Dumont. 4- 5 May 2006. GLM, Marseille, France*. 2006

11. Laurent U Perrinet, Jens Kremkow, Frédéric V Barthélemy, Guillaume S Masson and Frédéric Y Chavane. 'Input-output transformation in the visuo-oculomotor loop: modeling the ocular following response to center-surround stimulation in a probabilistic framework'. In: *Proceedings of the FENS Forum 2006*. 2006

10. Anna Montagnini, Pascal Mamassian, Laurent U Perrinet, Eric Castet and Guillaume S Masson. 'Bayesian modeling of dynamic motion integration'. In: *1ère conférence francophone NEUROsciences COMPutationnelles (NeuroComp)*. 2006. DOI: 10.1016/j.jphysparis.2007.10.013

9. Adrien Wohrer, Guillaume S Masson, Laurent U Perrinet, Pierre Kornprobst and Thierry Vieville. 'Contrast sensitivity adaptation in a virtual spiking retina and its adequation with mammalians retinas'. In: *European Conference on Visual Perception*. Ed. by Ricardo A. Carmona and Gustavo Linan-Cembrano. Vol. 35. 2006, p. 67

8. Laurent U Perrinet. 'Efficient Source Detection Using Integrate-and-Fire Neurons'. In: *International Conference on Artificial Neural Networks*. Ed. by David Hutchison et al. Vol. 3696. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer Berlin Heidelberg, 1st Jan. 2005. Chap. 27, pp. 167–172. ISBN: 978-3-540-28752-0. DOI: 10.1007/11550822_27

7. Laurent U Perrinet, Frédéric V Barthélemy, Eric Castet and Guillaume S Masson. 'Dynamics of motion representation in short-latency ocular following: A two-pathways Bayesian model'. In: *European Conference on Visual Perception*. Ed. by Ricardo A. Carmona and Gustavo Linan-Cembrano. Vol. 34. 2005, p. 38

6. Sylvain Fischer, Rafael Redondo, Laurent U Perrinet and Gabriel Cristóbal. 'Sparse Gabor wavelets by local operations'. In: *Microtechnologies for the New Millennium 2005*. Ed. by Ricardo A. Carmona and Gustavo Linan-Cembrano. Vol. 5839. Bioengineered and Bioinspired Systems II. Sevilla, Spain: SPIE, 29th June 2005, pp. 75–86. DOI: 10.1117/12.608403. URL: http://dx.doi.org/10.1117/12.608403

5. Sylvain Fischer, Rafael Redondo, Laurent U Perrinet and Gabriel Cristóbal. 'Efficient representation of natural images using local cooperation'. In: *European Conference on Visual Perception*. Ed. by Ricardo A. Carmona and Gustavo Linan-Cembrano. Vol. 34. 2005, p. 241

4. Rafael Redondo, Sylvain Fischer, Laurent U Perrinet and Gabriel Cristóbal. 'Modeling of simple cells through a sparse overcomplete gabor wavelet representation based on local inhibition and facilitation'. In: *European Conference on Visual Perception*.

Ed. by Ricardo A. Carmona and Gustavo Linan-Cembrano. Vol. 34. 1st Aug. 2005, p. 238

3. Laurent U Perrinet and Manuel Samuelides. 'Visual Strategies for Sparse Spike Coding'. In: *Actes de Neurosciences et Sciences de l'Ingenieur, L'Agelonde,* 2002

2. Laurent U Perrinet and Manuel Samuelides. 'Sparse Image Coding Using an Asynchronous Spiking Neural Network'. In: *Proceedings of ESANN*. 2002, pp. 313–8

1. Laurent U Perrinet and Manuel Samuelides. 'A generative model for Spike Time Dependent Hebbian Plasticity'. In: *Proceedings of DYNN*. 2000

7 Seminars and conferences with oral or poster presentations

82. Laurent U Perrinet. 'Introduction to Art & Neuroscience'. In: *NeuroDays 2026: The Art of Neuroscience*. Local chair. Aix-Marseille Université, Saint-Charles Campus, Marseille (France), 26th Nov. 2026. URL: https://neurodays2026.sciencesconf.org/resource/page/id/6

81. Laurent U Perrinet. 'Modelling memory to understand the brain'. In: *Forum Numerica seminar series, Academy of Excellence "Networks, Information and Digital society", Université Côte d'Azur*. Room 007, i3S laboratory, Bâtiment Euclide B, Sophia Antipolis (France), and online (Zoom), 9th Nov. 2026. URL: https://ds4h.univ-cotedazur.eu/research-and-labs/forum-numerica-seminar-series/forum-numerica-laurent-perrinet-cnrs-aix-marseille-university-modelling-memory-to-understand-the-brain

80. Laurent U Perrinet. 'Modelling memory to understand the brain'. In: *Seminar at Université Grenoble Alpes*. Room A6, Bâtiment Michel Dubois (or MSH amphitheatre), Saint-Martin-d'Hères, Grenoble (France), 3rd Nov. 2026

79. Laurent U Perrinet. 'Modelling working memory for neuroscience'. In: *Seminar at INT*. Marseille (France), 18th Sept. 2026. URL: https://laurentperrinet.github.io/publication/2026-09-18-int/

78. Laurent U Perrinet. 'Working Memory in Recurrent Spiking Neural Networks Using Heterogeneous Synaptic Delays'. In: *Seminar at CerCo*. Toulouse (France), 16th Apr. 2026. URL: https://laurentperrinet.github.io/publication/2026-04-16-cerco/

77. Laurent U Perrinet. 'Recréer des réseaux neuronaux pour améliorer la compréhension de notre cerveau'. In: *Séminaire en ligne biomimétisme et neurosciences*. En ligne, 10th Feb. 2026. URL: https://laurentperrinet.github.io/talk/2026-02-10-biomplus

76. Laurent U Perrinet. 'Al for visual processing and energy-efficient algorithms'. In: *AI and Neuroscience Day*. Marseille (France), 3rd Feb. 2026. URL: https://laurentperrinet.github.io/talk/2026-02-03-ai-and-neuroscience-day

75. Laurent U Perrinet. 'Neurosciences and sparsity'. In: *Séminaire au colloque du PEPR AI "Emergences" 2026*. Aussois (France), 29th Jan. 2026. URL: https://laurentperrinet.github.io/talk/2026-01-29-emergences

74. Laurent U Perrinet. 'A New Look for Convolutional Deep Networks'. In: *Montreal AI and Neuroscience conference, Dec 11-13th, 2025.*. Montréal (Québec), 12th Dec. 2025. URL: https://main2025.org

73. Laurent U Perrinet. 'A New Look for Convolutional Deep Networks'. In: *Symposium of the Neuroschool Master, Marseille, December 1 and 2 2025*. Marseille (France), 2nd Dec. 2025. URL: https://laurentperrinet.github.io/talk/2025-12-02-symposium-masters

72. Laurent U Perrinet. 'Mislocalization by Design: The Flash-Lag Effect as Prediction'. In: *Suresh Krishna's lab meeting at McGill*. Montréal (Québec), 16th Oct. 2025. URL: https://laurentperrinet.github.io/talk/2025-10-16-flash-lag-effect

71. Laurent U Perrinet. 'La vibration des apparences'. In: *Journées d'Ouverture Scientifique (JOS)*. Marseille (France), 18th Apr. 2025. URL: https://laurentperrinet.github.io/talk/2025-04-18-vibration-apparences

70. Laurent U Perrinet. 'Qu'est-ce que les Neurosciences peuvent apporter à l'Intelligence Artificielle ?' In: *Journée conviviale des anciens Sup'Aero*. Airbus Helicopters, Marignane (France), 14th Feb. 2025. URL: https://laurentperrinet.github.io/talk/2025-02-14-supaero

69. Laurent U Perrinet. 'When Cortical Neurons Talk Sideways: Beyond Feedforward Visual Processing'. In: *Séminaire Neuromathématiques*. Collège de France, Paris (France), 11th Feb. 2025. URL: https://laurentperrinet.github.io/talk/2025-02-11-neuromath

68. Laurent U Perrinet. 'How and why does retinotopy provides efficient vision'. In: *BRaIN Seminar, CRN and McGill Vision Research*. CRN seminar room, Montreal General Hospital, Livingston Hall, L7-140, Montreal (Canada), 8th Jan. 2025. URL: https://laurentperrinet.github.io/talk/2025-01-08-brain-seminar

67. Laurent U Perrinet. 'NeuroAI: interactions multiples entre Neurosciences et Intelligence artificielle'. In: *Journée Scientifique Biomimove 2024 : Action, Perception et Traitement*. HEXAGONE, 163 Avenue de Luminy - 13009 Marseille (France), 18th Nov. 2024. URL: https://laurentperrinet.github.io/talk/2024-11-18-biomim

66. Laurent U Perrinet. 'Robots aériens agiles bio-mimetiques pour le vol en conditions réelles: Enjeux pour l'IA embarqué'. In: *Journée Scientifique Enjeux pour l'IA embarqué*. Hexagone Balard, Paris (France), 9th Sept. 2024. URL: https://laurentperrinet.github.io/talk/2024-09-09-agileneurobot-anr

65. Laurent U Perrinet. 'Analyser de larges volumes de données neurobiologiques, vers une approche biomimétique'. In: *Séminaire au colloque du PEPR AI "Emergences" 2024*. Autrans (France), 27th Mar. 2024. URL: https://laurentperrinet.github.io/talk/2024-03-27-emergences

64. Laurent U Perrinet. 'Event-based vision'. In: *Seminar at UdeM's School of Optometry, UdeM*. Montréal (Canada), 5th Feb. 2024. URL: https://laurentperrinet.github.io/talk/2024-02-05-udem

63. Laurent U Perrinet. 'Event-based vision'. In: *Séminaire colloque BioComp 2023*. Banyuls-sur-Mer (France), 1st Dec. 2023. URL: https://laurentperrinet.github.io/talk/2023-12-01-biocomp

62. Adrien Fois and Laurent U Perrinet. 'Event-based vision'. In: *Journées sur l'apprentissage frugal (JRAF)*. Grenoble (France), 14th Dec. 2023. URL: https://laurentperrinet.github.io/talk/2023-12-14-jraf

61. Laurent U Perrinet. 'Accurate Detection of Spiking Motifs by Learning Heterogeneous Delays of a Spiking Neural Network'. In: *SNUFA: Spiking Neural networks as Universal Function Approximators*. Online, 7th Nov. 2023. URL: https://laurentperrinet.github.io/publication/perrinet-23-snufa/

60. Laurent U Perrinet. 'Accurate Detection of Spiking Motifs by Learning Heterogeneous Delays of a Spiking Neural Network'. In: *ICANN Special Session on Recent Advances in Spiking Neural Networks*. Heraklion (Greece), 27th Sept. 2023. URL: https://laurentperrinet.github.io/publication/perrinet-23-icann/

59. Laurent U Perrinet. 'Event-based vision'. In: *Séminaire à l'Institut Fresnel*. Marseille (France), 8th Sept. 2023. URL: https://laurentperrinet.github.io/talk/2023-09-08-fresnel

58. Laurent U Perrinet. 'Game theory and brain strategies'. In: *Atelier jeu et cerveau - M2 MASCO*. Marseille (France), 23rd Jan. 2023. URL: https://laurentperrinet.github.io/talk/2023-01-23-game-theory-and-the-brain

57. Antoine Grimaldi and Laurent U Perrinet. 'Learning heterogeneous delays of Spiking Neurons for motion detection'. In: *NeuroVision Workshop in conjunction with CVPR 2022*. New Orleans (virtual), 19th June 2022. URL: https://sites.google.com/uci.edu/neurovision2022/schedule

56. Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U Perrinet. 'Retinotopic mapping improves the reliability of image classification'. In: *NeuroVision Workshop in conjunction with CVPR 2022*. New Orleans (virtual), 19th June 2022. URL: https://sites.google.com/uci.edu/neurovision2022/schedule

55. Emmanuel Daucé and Laurent U Perrinet. 'Contributions of neuroscience to the detection and localization of objects in visual inputs'. In: *MIR Symposium 2022 bio-inspired and Marine Robotics*. SeaTech building, University of Toulon, 14th June 2022

54. Antoine Grimaldi and Laurent U Perrinet. 'Polychrony detection using heterogeneous delays'. In: *second CENTURI Scientific Day*. Marseille (France), 19th May 2022. URL: https://centuri-livingsystems.org/events/centuri-scientific-day-3/

53. Hugo Ladret and Laurent U Perrinet. 'Statistics of the sparse representations of natural images'. In: *2022 SIAM Conference on Imaging Science (IS22)*. 22nd Mar. 2022. URL: https://www.siam.org/conferences/cm/conference/is22

52. Laurent U Perrinet. 'Des illusions aux hallucinations visuelles: une porte sur la perception'. In: *Neurocercle : Découvrir les neurosciences à Grenoble*. Grenoble (France), 12th Jan. 2022. URL: https://laurentperrinet.github.io/talk/2022-01-12-neuro-cercle

51. Hugo Ladret, Nelson Cortes, Lamyae Ikan, Frédéric Chavane, Christian Casanova and Laurent U Perrinet. 'Dynamical processing of orientation precision in the primary visual cortex'. In: *DynamicsDays - XL*. 27th Aug. 2021

50. Laurent U Perrinet and Angelo Franciosini. 'Pooling in a predictive model of V1 explains functional and structural diversity across species'. In: *Society for Mathematical Biology * 2021*. Everywhere (World), 15th June 2021. URL: http://schedule.smb2021.org/MS05/

49. Hugo Ladret, Nelson Cortes, Lamyae Ikan, Frédéric Chavane, Christian Casanova and Laurent U Perrinet. 'Dynamical processing of orientation precision in the primary visual cortex'. In: *NeuroFrance 2021*. 20th May 2021. URL: https://virtualnf21.neurosciences.asso.fr/meetings/virtual/m3LQJKS4fbaWkg7rE

48. Laurent U Perrinet. 'Understanding natural vision using deep predictive coding'. In: *Séminaire à l'Institut de Recherche sur les Phénomènes Hors Équilibre*. Marseille (France), 25th Sept. 2020. URL: https://laurentperrinet.github.io/talk/2020-09-25-irphe

47. Emmanuel Daucé and Laurent Perrinet. 'Visual search as active inference'. In: *IWAI 2020*. Ghent (Belgium), gone virtual, 14th Sept. 2020. DOI: 10.1007/978-3-030-64919-7_17. URL: https://whova.com/embedded/subsession/ecmlp_202009/1215095/1215123/

46. Laurent U Perrinet. 'From the retina to action: Understanding visual processing'. In: *Master Neurosciences et Sciences Cognitives*. Marseille (France), 3rd Apr. 2020. URL: https://laurentperrinet.github.io/talk/2020-04-ue-neurosciences-computationnelles

45. Laurent U Perrinet. 'Des illusions aux hallucinations visuelles: une porte sur la perception'. In: *Cinéma et sciences : rencontre avec les élèves du lycée des métiers*. Lycée Professionnel Domaine Eguille, Vedène (France), 20th Jan. 2020. URL: https://laurentperrinet.github.io/talk/2020-01-20-atelier-sciences-cinema

44. Emmanuel Daucé, Pierre Albigès and Laurent U Perrinet. 'Learning where to look: a foveated visuomotor control model'. In: *Annual Computational Neuroscience Meeting: CNS*2019 Barcelona, Spain*. 2019. URL: https://bmcneurosci.biomedcentral.com/articles/10.1186/s12868-019-0538-0#Sec73

43. Laurent U Perrinet. 'Should I stay or should I go? Humans adapt to the volatility of visual motion properties, and know about it'. In: *Colloque international de la Société Française des Neurosciences 2019*. Marseille (France), 23rd May 2019. URL: https://laurentperrinet.github.io/talk/2019-05-23-neurofrance

42. Laurent U Perrinet. 'Des illusions aux hallucinations visuelles: une porte sur la perception'. In: *JNLF 2019, Revue Neurologique, Volume 175, Supplement 1, Page S165*. Lille, France, 2019. DOI: 10.1016/j.neurol.2019.01.031. URL: https://www.em-consulte.com/e-tap/1283936/ftafhrsrftfxjyjaaumj

41. Laurent U Perrinet. 'Should I stay or should I go? Adaption of human observers to the volatility of visual inputs'. In: *CausaL Kick-off*. INT, Marseille (France), 5th Apr. 2019. URL: https://laurentperrinet.github.io/talk/2019-04-05-bbcp-causal-kickoff

40. Victor Boutin, Angelo Franciosini and Laurent U Perrinet. 'From the retina to action: Predictive processing in the visual system'. In: *HDR Robin Baurès, Toulouse (France)*. Toulouse (France), 25th Mar. 2019. URL: https://laurentperrinet.github.io/2019-03-25_HDR_RobinBaures

39. Laurent U Perrinet. 'Should I stay or should I go? Adaption of human observers to the volatility of visual inputs'. In: *LACONEU 2019: 5th Latin-American Summer School in Computational Neuroscience*. Valparaiso (Chile), 2019. URL: https://laurentperrinet.github.io/talk/2019-01-18-laconeu/

38. Laurent U Perrinet, Chloé Pasturel and Anna Montagnini. 'Principles and psychophysics of Active Inference in anticipating a dynamic, switching probabilistic bias'. In: *Probabilities and Optimal Inference to Understand the Brain*. INT, Marseille (France), 2018. URL: https://laurentperrinet.github.io/talk/2018-04-05-bcp-talk/

37. Laurent U Perrinet. 'Back to the present: how neurons deal with delays'. In: *Workshop on Computational Neuroscience "New trends and challenges for 2030"*. Valparaiso (Chile), 2017. URL: https://laurentperrinet.github.io/talk/2017-01-18-laconeu/

36. Laurent U Perrinet. 'Back to the present: dealing with delays in biological and neuromorphic systems'. In: *Workshop on Computational Neuroscience entitled "Neuromorphic Event-based Compound Eyes and Vision"*. Telluride, CO, 2017. URL: https://laurentperrinet.github.io/talk/2017-06-28-telluride

35. Laurent U Perrinet. 'Modelling the dynamics of cognitive processes: from the Bayesian brain to particles'. In: *Summer School: PDE and Probability for Life Sciences*. CIRM, Marseille, 2016. URL: https://laurentperrinet.github.io/talk/2016-07-07-edp-proba/

34. Laurent U Perrinet. 'Eye movements as a model for active inference'. In: *Lyon Active inference Workshop (LAW)*. Lyon, France, 2016. URL: https://laurentperrinet.github.io/talk/2016-10-13-law/

33. Laurent U Perrinet. 'The flash-lag effect as a motion-based predictive shift'. In: *Workshop SIGMA'2016: Signal, Image, Geometry, Modelling, Approximation*. CIRM, 2016. URL: https://laurentperrinet.github.io/talk/2016-11-03-sigma/

32. Lionel Fillatre, Michel Barlaud and Laurent U Perrinet. 'Categorization of microscopy images using a biologically inspired edge co-occurrences descriptor'. In: *EUVIP Session 7: Biologically Inspired Computer Vision (Special Session)*. Ecole Centrale Marseille, 2016. URL: https://laurentperrinet.github.io/talk/2016-10-26-fillatre-barlaud-perrinet-16-euvip/

31. Laurent U Perrinet. 'Biologically-inspired characterization of sparseness in natural images'. In: *EUVIP Session 7: Biologically Inspired Computer Vision (Special Session)*. Ecole Centrale Marseille, 2016. URL: https://laurentperrinet.github.io/publication/perrinet-16-euvip/

30. Jean-Bernard Damasse, Laurent U Perrinet, Jérémie Jozefowiez, Laurent Madelain and Anna Montagnini. 'Reinforcement contingencies modulate anticipatory smooth eye movements'. In: *GDR Vision, Toulouse, Nov 3rd, 2016*. 2016. URL: https://laurentperrinet.github.io/talk/2016-11-03-gdr/

29. Laurent U Perrinet. 'Motion-based prediction with neuromorphic hardware'. In: *Charla*. Universidad Tecnica Federico Santa Maria, Valparaiso (Chile), 2015. URL: https://laurentperrinet.github.io/talk/2015-11-05-chile/

28. Laurent U Perrinet. 'Motion-based prediction with neuromorphic hardware'. In: *First GDR BioComp workshop*. Saint-Paul de Vence, 2015. URL: https://laurentperrinet.github.io/talk/2015-10-07-gdr-bio-comp/

27. Laurent U Perrinet. 'Axonal delays and on-time control of eye movements'. In: *Marseille INT Fest, January 10th, 2014*. 2014. URL: https://laurentperrinet.github.io/talk/2014-01-10-int-fest/

26. Laurent U Perrinet, Bernhard A Kaplan, Mina A Khoei, Anders Lansner and Guillaume S Masson. 'WP5 - Demo 1.3 : Spiking model of motion-based prediction'. In: *4th BrainScaleS Plenary meeting*. Manchester (UK), 2014. URL: https://laurentperrinet.github.io/talk/2014-03-20-manchester/

25. Bernhard A Kaplan, Mina A Khoei, Anders Lansner and Laurent U Perrinet. 'Signature of an anticipatory response in area V1 as modeled by a probabilistic model and a spiking neural network'. In: *2014 International Joint Conference on Neural Networks (IJCNN)*. Beijing, China: IEEE, 2014, pp. 3205–3212. ISBN: 978-1-4799-1484-5. DOI: 10.1109/IJCNN.2014.6889847. URL: https://laurentperrinet.github.io/talk/2014-04-25-kaplan-beijing/

24. Laurent U Perrinet. 'Why methods and tools are the key to artificial brain-like systems'. In: *3rd BrainScaleS Plenary Meeting - Friday, March 21st, 2013*. Location: INT, Marseille. 2013. URL: https://laurentperrinet.github.io/talk/2013-03-21-marseille/

23. Laurent U Perrinet, David Fitzpatrick and James A Bednar. 'Edge co-occurrences and categorizing natural images'. In: *CerCo 20th anniversary*. CerCo, Toulouse, 2013. URL: https://laurentperrinet.github.io/talk/2013-07-05-cerco/

22. Bernhard A Kaplan and Laurent U Perrinet. 'Demo 1, Task4: Implementation of models showing emergence of cortical fields and maps'. In: *Demo 1-3: Apparent Motion in V1/ MT/MST: Neural Implementation of Probabilistic Approaches*. 2013. URL: https://laurentperrinet.github.io/talk/2013-11-26-brain-scales-demos/

21. Laurent U Perrinet. 'Motion-based prediction is sufficient to solve the aperture problem'. In: *Vision@UCL seminar*. Malet Place Eng Bldg 1.03 (first floor)., 2012. URL: https://laurentperrinet.github.io/talk/2012-01-12-vision-at-ucl/

20. Laurent U Perrinet, David Fitzpatrick and James A Bednar. 'Edge statistics in natural images versus laboratory animal environments: implications for understanding lateral connectivity in V1'. In: *A seminar from the Institute for Adaptive and Neural Computation (ANC)*. Room IF 4.31/4.33, Institute for Adaptive and Neural Computation (ANC) at the University of Edinburgh, 2012. URL: https://laurentperrinet.github.io/talk/2012-01-24-edinburgh/

19. Laurent U Perrinet. 'Grabbing, tracking and sniffing as models for motion detection and eye movements'. In: *Brain meeting at FIL, London - Friday, January 27th, 2012*. 2012. URL: https://laurentperrinet.github.io/talk/2012-01-27-fil/

18. Laurent U Perrinet. 'MotionClouds: Model-based stimulus synthesis of natural-like random textures for the study of motion perception'. In: *Second BrainScaleS plenary Meeting - WP4*. Forschungszentrum Jülich, 2012. URL: https://laurentperrinet.github.io/talk/2012-03-22-juelich/

17. Laurent U Perrinet. 'Apparent motion in V1 - Probabilistic approaches'. In: *Second BrainScaleS plenary Meeting - WP5*. Forschungszentrum Jülich, 2012. URL: https://laurentperrinet.github.io/talk/2012-03-23-juelich/

16. Laurent U Perrinet, David Fitzpatrick and James A Bednar. 'Edge statistics in natural images versus laboratory animal environments: implications for understanding lateral connectivity in V1'. In: *iTWIST '12 workshop*. 2012. URL: https://laurentperrinet.github.io/talk/2012-05-10-itwist/

15. Laurent U Perrinet. 'Propriétés émergentes d'un modèle de prédiction probabiliste utilisant un champ neural'. In: *Atelier Neurosciences Computationnelles, 2-3 Juillet 2011 Khemisset, Maroc*. 2011. URL: https://laurentperrinet.github.io/talk/2011-07-02-neuro-med-talk/

14. Laurent U Perrinet, David Fitzpatrick and James A Bednar. 'Edge statistics in natural images versus laboratory animal environments: implications for understanding lateral connectivity in V1'. In: *Proceedings of SfN, 2011*. Porquerolles la Perle des Iles d'Or - Var (France), 2011. URL: https://laurentperrinet.github.io/talk/2011-09-28-ermites/

13. Laurent U Perrinet. 'Demo 1, Task4: Implementation of models showing emergence of cortical fields and maps'. In: *Using the ESS + Neuromorphic hardware Workshop*. TU Dresden, Germany, 2011. URL: https://laurentperrinet.github.io/talk/2011-10-05-brain-scales-ess/

12. Laurent U Perrinet, David Fitzpatrick and James A Bednar. 'Edge statistics in natural images versus laboratory animal environments: implications for understanding lateral connectivity in V1'. In: *Society for Neuroscience Abstracts*. Ed. by www.sfn.org Society for Neuroscience. Program No. 530.04. Washington, DC, 2011. URL: https://laurentperrinet.github.io/talk/2011-11-15-sfn/

11. Laurent U Perrinet and Guillaume S Masson. 'Models of low-level vision: linking probabilistic models and neural masses'. In: 2010. URL: https://laurentperrinet.github.io/talk/2010-01-08-facets/

10. Laurent U Perrinet. 'Probabilistic models of the low-level visual system: the role of prediction in detecting motion'. In: *LADISLAV TAUC and GDR MSPC NEUROSCIENCES CONFERENCE, From Mathematical Image Analysis to Neurogeometry of the Brain*. 2010. URL: https://laurentperrinet.github.io/talk/2010-12-17-tauc-talk/

9. Laurent U Perrinet and Guillaume S Masson. 'Decoding low-level neural information to track visual motion'. In: 2009. URL: https://laurentperrinet.github.io/talk/2008-04-01-incm/

8. Jens Kremkow, Laurent U Perrinet, Cyril Monier, Yves Frégnac, Guillaume S Masson and Ad M Aertsen. 'Control of the temporal interplay between excitation and inhibition by the statistics of visual input'. In: *Eighteenth Annual Computational Neuroscience Meeting: CNS*2009 Berlin, Germany. 18–23 July 2009*. 2009, Oral presentation, 10(Suppl 1):O21. DOI: 10.1186/1471-2202-10-s1-o21

7. Laurent U Perrinet, Alexandre Reynaud, Frédéric Y Chavane and Guillaume S Masson. 'Reading out the dynamics of lateral interactions in the primary visual cortex from VSD data'. In: *Macroscopic aspects of neuronal activity: "Macroscopic models, LFP models and VSD models" a FACETS workshop in Marseille, Nov. 30th /Dec. 1st*. 2009. URL: https://laurentperrinet.github.io/talk/2009-11-30-vss/

6. Laurent U Perrinet. 'Modeling of spikes, sparseness and adaptation in the primary visual cortex: applications to imaging'. In: *Prisma workshop, Toledo (Spain), February 7, 2008*. 2008. URL: https://laurentperrinet.github.io/talk/2008-02-01-toledo/

5. Laurent U Perrinet. 'From neural activity to behavior: computational neuroscience as a synthetic approach for understanding the neural code.' In: *Séminaires de l'INCM, April 11th, 2008*. 2008. URL: https://laurentperrinet.github.io/talk/2008-04-01-incm/

4. Laurent U Perrinet and Guillaume S Masson. 'Decoding the population dynamics underlying ocular following response using a probabilistic framework'. In: 2008. URL: https://laurentperrinet.github.io/talk/2008-06-01-ulm/

3. Laurent U Perrinet. 'Neural Codes for Adaptive Sparse Representations of Natural Images'. In: *Mathematical image processing meeting (Marseille, France) September 5*. 2007. URL: https://laurentperrinet.github.io/talk/2007-09-01-mipm

2. Laurent U Perrinet. 'What efficient code for adaptive spiking representations?' In: *The Rank Prize Funds, Mini-Symposium on Representations of the Visual World in the Brain*. 2007. URL: https://laurentperrinet.github.io/talk/2007-12-01-rankprize/

1. Laurent U Perrinet, Frédéric V Barthélemy and Guillaume S Masson. 'Input-output transformation in the visuo-oculomotor loop: modeling the ocular following response to center-surround stimulation in a probabilistic framework'. In: *1ère conférence francophone NEUROsciences COMPutationnelles - NeuroComp*. 2006. URL: https://laurentperrinet.github.io/publication/perrinet-06-neurocomp/

8 Courses and science outreach activities

43. Laurent U Perrinet. 'Dans l'intelligence du regard: l'art révèle la diversité de notre vision'. In: *Forum des Sciences Cognitives 2026*. Paris (France), 11th Apr. 2026.

URL: https://laurentperrinet.github.io/talk/2026-04-11-intelligence-du-regard

42. Laurent U Perrinet. 'Spiking neural nets'. In: *NeuroSchool PhD Program in Neuroscience*. Marseille (France), 24th Mar. 2026. URL: https://laurentperrinet.github.io/talk/2026-03-24_phd-program-spiking-neural-nets

41. Laurent U Perrinet. 'Artificial neural networks and machine learning applied to the understanding of biological vision'. In: *Master 1 Neuroscience, UE Natural Cognition, Artificial Cognition.* Marseille (France), 5th Mar. 2026. URL: https://laurentperrinet.github.io/talk/2026-03-05-ue-natural-cognition

40. Laurent Perrinet and Etienne Rey. 'A New Look for Convolutional Deep Networks'. In: *Lab Tour for Art - Perception Course, January 19th, 2026*. Marseille (France), 2026

39. Laurent U Perrinet. 'Conférence immersive : Ouvrez grand les yeux'. In: *Festival EXPLORE organisé par Aix Marseille Université et la DR12 du CNRS.*. Marseille (France), 12th June 2025. URL: https://laurentperrinet.github.io/talk/2025-06-12_explore-conference-immersive

38. Laurent U Perrinet. 'Ciné-Sciences : Ouvrez grand les yeux'. In: *Festival EXPLORE organisé par Aix Marseille Université et la DR12 du CNRS.*. Marseille (France), 10th June 2025. URL: https://laurentperrinet.github.io/talk/2025-06-10_explore-cine-sciences

37. Laurent U Perrinet. 'Artificial neural networks and machine learning applied to the understanding of biological vision'. In: *Master M4NC de l'institut NeuroMod, cours Prospective Innovation and Research.* Sophia-Antipolis (France), 26th May 2025. URL: https://laurentperrinet.github.io/talk/2025-05-26-master-m-4-nc

36. Laurent U Perrinet. 'Sparse representations'. In: *NeuroSchool PhD Program in Neuroscience*. Marseille (France), 11th Mar. 2025. URL: https://laurentperrinet.github.io/talk/2025-03-11-phd-program-sparse-representations

35. Laurent U Perrinet and Hugo Ladret. 'Le mystère de la Joconde élucidé par les neurosciences:' in: *Cerveau & Psycho* Nř 168.8 (25th Aug. 2024), pp. 30–36. ISSN: 1639-6936. DOI: 10.3917/cerpsy.168.0030. URL: https://www.cerveauetpsycho.fr/sd/neurobiologie/le-mystere-de-la-joconde-elucide-par-les-neurosciences-26605.php

34. Laurent U Perrinet. 'Chats, mouches, humains : comment la vision a évolué en de multiples facettes'. In: *The Conversation* (23rd Feb. 2024). URL: https://theconversation.com/chats-mouches-humains-comment-la-vision-a-evolue-en-de-multiples-facettes-220083

33. Laurent U Perrinet. 'Formes et perception'. In: *Vasarely, d'un art programmatique au numérique*. 2023. ISBN: 978-88-366-4958-7. URL: https://laurentperrinet.github.io/2023-01-31_formes-et-perception

32. Laurent U Perrinet. 'Artificial neural networks and machine learning applied to the understanding of biological vision'. In: *Master M4NC de l'institut NeuroMod, cours Prospective Innovation and Research.* Sophia-Antipolis (France), 13th May 2024. URL: https://laurentperrinet.github.io/talk/2024-05-13-master-m-4-nc

31. Laurent U Perrinet. 'Sparse representations'. In: *NeuroSchool PhD Program in Neuroscience*. Marseille (France), 17th Apr. 2024. URL: https://laurentperrinet.github.io/talk/2024-04-17-phd-program-sparse-representations

30. Laurent U Perrinet. 'Artificial neural networks applied to the understanding of biological vision'. In: *Master 1 Neurosciences et Sciences Cognitives*. Marseille (France), 10th Apr. 2024. URL: https://laurentperrinet.github.io/talk/2024-04-10-ue-neurosciences-computationnelles

29. Laurent U Perrinet. 'Interactions between machine learning, artificial neural networks and our understanding of biological vision'. In: *NeuroSchool PhD Program in Neuroscience*. Marseille (France), 10th May 2023. URL: https://laurentperrinet.github.io/talk/2023-05-10-phd-program-neurosciences-computationnelles/

28. Laurent U Perrinet. 'Artificial neural networks and machine learning applied to the understanding of biological vision'. In: *Master 1 Neurosciences et Sciences Cognitives*. Marseille (France), 5th Apr. 2023. URL: https://laurentperrinet.github.io/talk/2023-04-05-ue-neurosciences-computationnelles/

27. Laurent U Perrinet. 'Artificial neural networks and machine learning applied to the understanding of biological vision'. In: *Master M4NC de l'institut NeuroMod, cours Prospective Innovation and Research.* Sophia-Antipolis (France), 3rd Apr. 2023. URL: https://laurentperrinet.github.io/talk/2023-04-03-master-m-4-nc/

26. Laurent U Perrinet. 'Réseaux de neurones artificiels et apprentissage machine appliqués à la compréhension de la vision'. In: *Master 1 Neurosciences et Sciences Cognitives*. Marseille (France), 23rd Mar. 2022. URL: https://laurentperrinet.github.io/talk/2022-03-23-ue-neurosciences-computationnelles

25. Laurent U Perrinet. 'Le jeu du cerveau et du hasard'. In: *The Conversation* (2021). DOI: 10.64628/aak.erjttwvdp. URL: https://laurentperrinet.github.io/publication/perrinet-21-hasard/

24. Laurent U Perrinet. 'Temps et cerveau : comment notre perception nous fait voyager dans le temps'. In: *The Conversation* (2019). DOI: 10.64628/aak.ktksa6fr5. URL: https://theconversation.com/temps-et-cerveau-comment-notre-perception-nous-fait-voyager-dans-le-temps-127567

23. Laurent U Perrinet. 'Illusions et hallucinations visuelles : une porte sur la perception'. In: *The Conversation* (2019). DOI: 10.64628/aak.pwc9p7tpd. URL: https://theconversation.com/illusions-et-hallucinations-visuelles-une-porte-sur-la-perception-117389

22. Laurent U Perrinet. 'Rencontre avec les collégiens marseillais'. In: *Cinéma et sciences : rencontre avec les collégiens marseillais*. Marseille, France, 2019. URL: https://laurentperrinet.github.io/talk/2019-01-10-polly-maggoo/

21. Laurent U Perrinet. 'Modelling spiking neural networks using Brian, Nest and pyNN'. in: *LACONEU 2019: 5th Latin-American Summer School in Computational Neuroscience*. Valparaiso (Chile), 2019. URL: https://laurentperrinet.github.io/talk/2019-01-14-laconeu/

20. Laurent U Perrinet. 'Efficient coding of visual information in neural computations'. In: *LACONEU 2019: 5th Latin-American Summer School in Computational Neuroscience*. Valparaiso (Chile), 2019. URL: https://laurentperrinet.github.io/talk/2019-01-16-laconeu/

19. Laurent U Perrinet. 'Role of dynamics in neural computations underlying visual processing'. In: *LACONEU 2019: 5th Latin-American Summer School in Computational Neuroscience*. Valparaiso (Chile), 2019. URL: https://laurentperrinet.github.io/talk/2019-01-17-laconeu/

18. Laurent U Perrinet. 'From the retina to action: Understanding visual processing'. In: *Licence Sciences et Humanité*. Marseille (France), 3rd Apr. 2019. URL: https://laurentperrinet.github.io/talk/2019-04-03-a-course-on-vision-and-modelization

17. Laurent U Perrinet and Etienne Rey. 'Expériences autour de la perception de la forme en art et science'. In: *Meetup Art et Neurosciences, Association NeuroNautes*. Salle des voutes campus Saint Charles, 2018. URL: https://laurentperrinet.github.io/talk/2018-01-25-meetup-neuronautes/

16. Laurent U Perrinet, Chloé Pasturel and Anna Montagnini. 'Estimating and anticipating a dynamic probabilistic bias in visual motion direction'. In: *Visual motion Fest - Invibe Team – INT / Marseille February 1 and 2, 2018*. 2018. URL: https://laurentperrinet.github.io/talk/2018-02-01-bcp-invibe-fest/

15. Laurent U Perrinet. 'Probabilities, Bayes and the Free-energy principle'. In: *Course in Computational Neuroscience @ PhD program*. INT, Marseille, 2018. URL: https://laurentperrinet.github.io/talk/2018-03-26-cours-neuro-comp-fep/

14. Laurent U Perrinet. 'Intervention fête de la science 2018'. In: *FÊTE DE LA SCIENCE 2018 : Alcazar / MERLAN*. Marseille, France, 2018. URL: https://laurentperrinet.github.io/talk/2018-10-10-polly-maggoo/

13. Laurent U Perrinet. 'La modélisation biomorphique de la perception visuelle'. In: *in 'La modélisation de la genèse physico-mathématique du vivant' / BIOMORPHISME ET CREATION ARTISTIQUE Session 3*. Marseille, France, 2018. URL: https://laurentperrinet.github.io/sciblog/files/2018-10-11_BioMorphisme.html

12. Laurent U Perrinet. 'Tutorial: Sparse optimization in neural computations'. In: *LACONEU 2017: 4th Latin-American Summer School in Computational Neuroscience*. Valparaiso (Chile), 2017. URL: https://laurentperrinet.github.io/talk/2017-01-19-laconeu/

11. Laurent U Perrinet. 'Tutorial: Active inference for eye movements: Bayesian methods, neural inference, dynamics'. In: *LACONEU 2017: 4th Latin-American Summer School in Computational Neuroscience*. Valparaiso (Chile), 2017. URL: https://laurentperrinet.github.io/talk/2017-01-20-laconeu/

10. Laurent U Perrinet. 'Tutorial on predictive coding'. In: *Telluride Neuromorphic Workshop, Workgroup on Compound Eyes and Event-based Vision*. Telluride, CO, 2017. URL: https://laurentperrinet.github.io/talk/2017-06-30-telluride

9. Laurent U Perrinet. 'What dynamic neural codes for efficient visual processing'. In: *Colloque : "CODAGES ET REPRESENTATIONS", MASTER DE NEUROSCIENCES 2ème année; Comité d'organisation: Francesca SARGOLINI, Christian Bénar, Paolo GUBELLINI, Christian GESTREAU*. Aix-Marseille Université, Campus Saint-Charles, Salle des voûtes, 2017. URL: https://laurentperrinet.github.io/talk/2017-11-15-colloque-master/

8. Laurent U Perrinet. 'Participation au jury'. In: *Festival Interférences - Cinéma Documentaire et Débat Public*. Lyon, France, 2017. URL: https://laurentperrinet.github.io/talk/2017-11-17-festival-interferences/

7. Victor Boutin, Franck Ruffier and Laurent U Perrinet. 'Unsupervised learning applied to robotic vision'. In: IMERA (Aix-Marseille Université), 2017. URL: https://laurentperrinet.github.io/talk/2017-11-24-neurosciences-robotique/

6. Laurent U Perrinet. 'Les illusions visuelles, un révélateur du fonctionnement de notre cerveau'. In: *Cinésciences, collège Clair Soleil (Marseille)*. Marseille, France, 2016. URL: https://laurentperrinet.github.io/sciblog/files/2016-04-25_pollymagoo/

5. Laurent U Perrinet. 'Les illusions visuelles, un révélateur du fonctionnement de notre cerveau'. In: *Cycle de conférences "Tous connectés", Bibliothèque de Méjanes*. Marseille, France, 2016. URL: https://laurentperrinet.github.io/sciblog/files/2016-04-28_mejanes/

4. Laurent U Perrinet. 'Participation au jury et entretien avec Clara Delmon'. In: *Rencontres Internationales Sciences Et Cinémas*. Marseille, France, 2016. URL: https://laurentperrinet.github.io/talk/2016-11-20-polly-maggoo/

3. Laurent U Perrinet. 'Diffraction monochromatique, spectre audiographique'. In: *intervention autour du vernissage de "Diffraction monochromatique, spectre audiographique" d'Etienne Rey*. Aix-enProvence (France), 2010. URL: https://laurentperrinet.github.io/talk/2010-04-14-ondes-paralleles/

2. Laurent U Perrinet. 'Qui créera le premier ordinateur intelligent?' In: *DocSciences* 13 (20th June 2011). URL: https://interstices.info/qui-creera-le-premier-ordinateur-intelligent/

1. Laurent U Perrinet and Thierry Viéville. 'Peut-on parler d'intelligence mécanique?' In: *Cycle de conférences organisé par l'Association Science Technologie Société - PACA ayant pour thème cette année : "Biologie et civilisation : les chemins de l'intelligence"*. Marseille, France, 2009. URL: https://laurentperrinet.github.io/talk/2009-11-24-intelligence-mecanique/